香港科技大学(广州)
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY (GUANGZHOU)

UCUG 1903 Signals and Information Technology



2026-27 Fall Term

Lecture 1 Introduction to Signals and Systems

Dr. Yeyu TONG

yeyutong@hkust-gz.edu.cn

Acknowledgement



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PhD in Electronic and Computer Engineering
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Assistant Professor of Practice in Engineering Education
Pillar of STEM Education



Shijian GAO
高诗简

PhD in Electrical Engineering
University of Minnesota - Twin Cities Campus, 2022

Assistant Professor
Thrust of Internet of Things

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Course Information

- **Instructor**: Dr. Yeyu TONG
- **Email**: yeyutong@hkust-gz.edu.cn
- **Lecture Time & Location**: Wednesday 16:30-18:20, W4-101
- **Tutorial Time & Location**: Thursday 12:00-12:50, E1-201

(**Please bring your laptop and pre-install MATLAB using the provided guide.**)

- **Office Hour**: Thursday 14:00-15:00, W3 508
- **Teaching Assistants:** Mr. Xianyi FENG
- **Email:** xfeng262@connect.hkust-gz.edu.cn

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Weekly Syllabus

Week	Dates	Lecture
Week1	Sep-2	Introduction to Signals and Systems;
Week2	Sep-9	Sound Signal, Frequency and Harmonics, Signals as Sum of Sine Waves
Week3	Sep-16	Spectrum - Representation of Signals in the Frequency; Systems as Filters of Signals
Week4	Sep-23	Benefits of Digitization, Bits and Bytes
Week5	Sep-30	Introduction to Analog to Digital Conversion, Quantization
Week6	Oct-14	Claude Shannon and Information Theory, Introduction to Source Coding
Week7	Oct-21	Huffman Code and Moving Picture Experts Group (MPEG)
	TBA	**Midterm**
Week8	Oct-28	Error Detection Codes, Error Correcting Code
Week9	Nov-04	Channel Capacity, Introduction to Wireless Communications
Week10	Nov-11	Cellular Network Basics Frequency Translation
Week11	Nov-18	Multiple Access Technologies
Week12	Nov-25	Introduction to Internet Protocol and Packet Switching
Week13	Dec-02	Project Presentation
By 25 Dec	TBA	Final Exam

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Power of Information Technology in a Single Handheld Device



❑ Smart Scan

(Objects/Fingerprint/Iris/Face Recognition)

❑ 5G Mobile

(***Mobile communications***)

❑ Apple/Huawei Pay

(Near Field Communications)



❑ Voice Assistant

(Digital Audio Recognition/Matching)





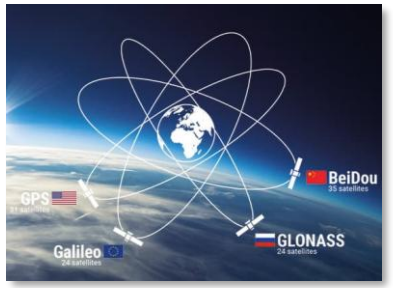
❑ Satellite Calls

(Satellite Mobile Communication System, Microchip Technology, Signal Amplification)



❑ Camera

(Digital Image and Video Processing)

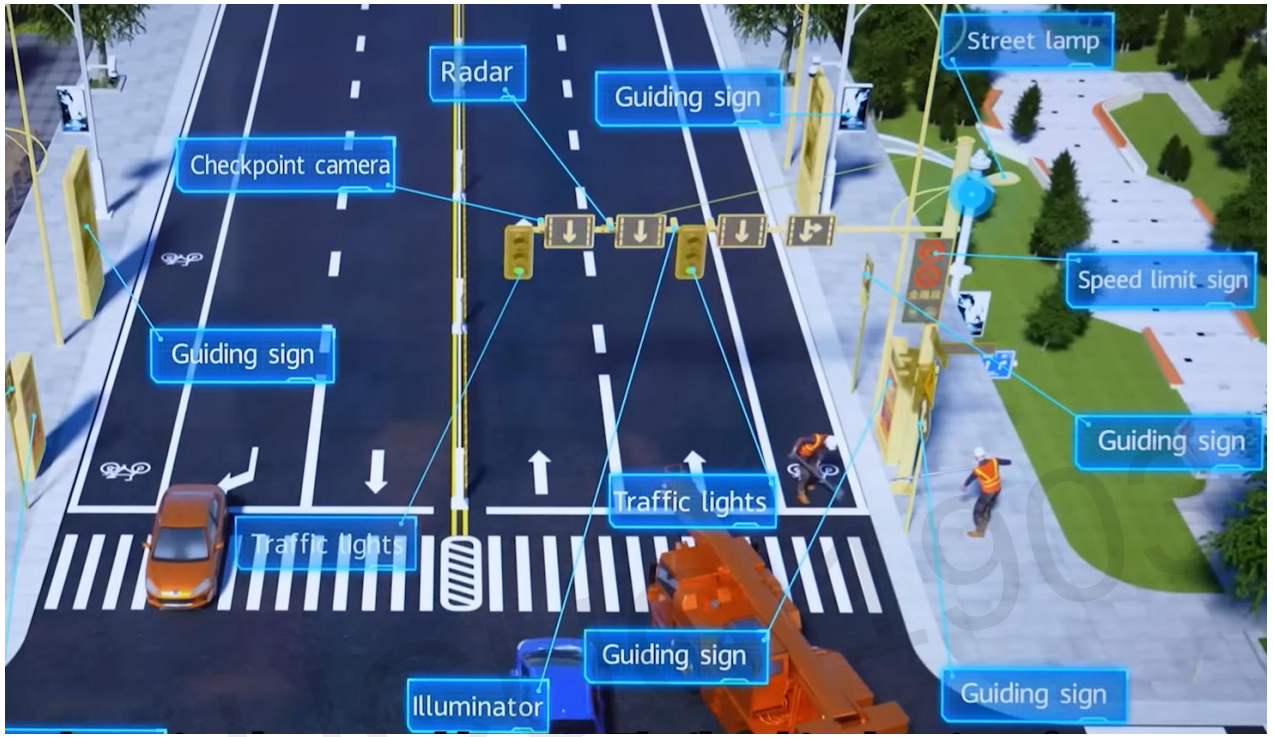


❑ Global Navigation Satellite System

(Satellite Positioning and Navigation Technology)

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Smart Management System





Signal and Information Technology

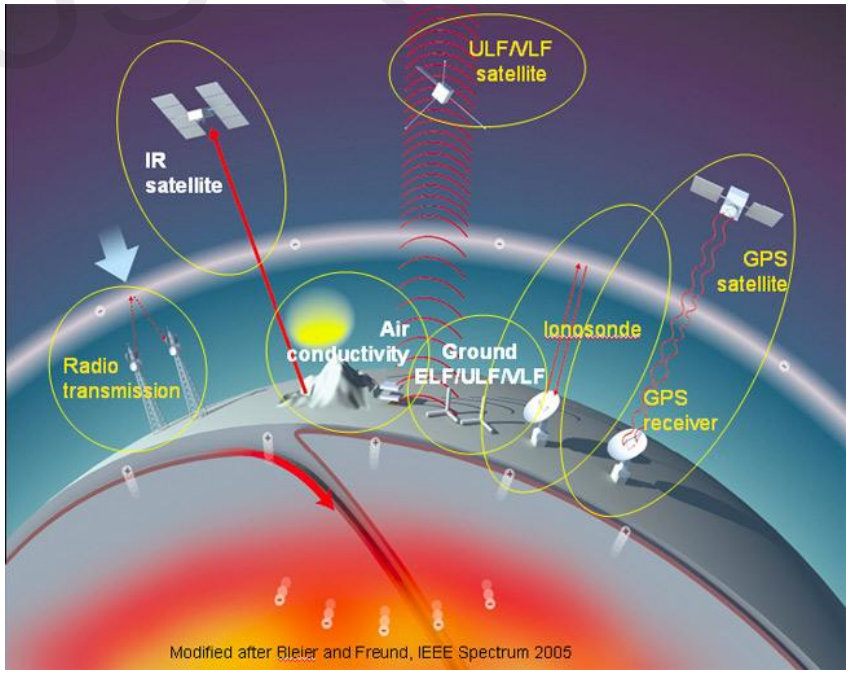
- Signals and signal processing are pervasive in our world and enable many of our everyday experiences.
- Information technology has made the education process more effective and productive.
- Information technologies (IT) are drivers of productivity and the economies.
- Information technologies often enable scientific and technological progresses in many disciplines (e.g. broadband Internet enables tele-medicine; wireless sensor networks enables earthquake and tsunami detection).
- Huge investment and job market $$$.











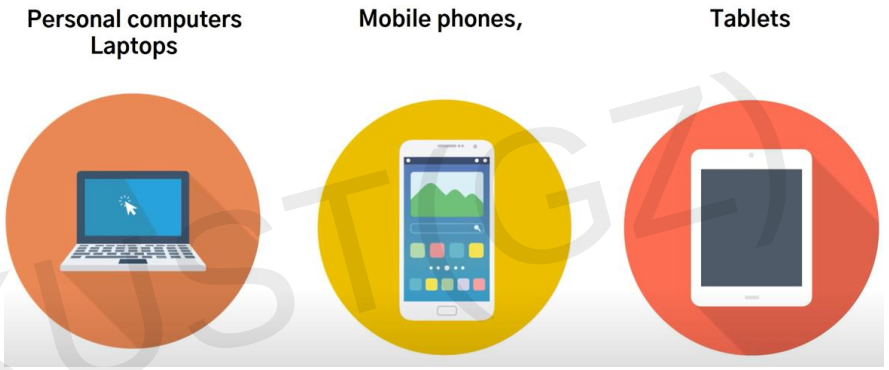
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What is Information Technology?

- **Information Technology Association of America (ITAA):** "the study, design, development, implementation, support or management of computer-based information systems, particularly software applications and computer hardware." IT deals with the use of electronic computers and computer software to convert, store, protect, process, transmit and retrieve information, securely.

- Sometimes the term “Information Technology” is also used to referred to everything that is high-tech.

Hardware



Software

Software systems

Applications











Pictures are from https://commons.wikimedia.org/wiki/File:HarmonyOS.jpg, https://www.youtube.com/watch?v=XZrckLYqdys

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Course Materials

- No textbook.
- There will be lecture notes, summary notes, some *optional* reading articles, and video tutorials from the Internet.
- Tutorial, midterm and final exams will be based on the lecture notes and summary notes.

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Intended Learning Outcomes

- On successful completion of this course, you will be able to
  - Describe the key technological developments of signals and information technology;
  - Identify the fundamental principles related to signals and information technology;
  - Solve simple engineering problems related to signals and information technology, such as audio signal processing and modulation, using software tools;
  - Create an interactive presentation on state-of-the-art signals and information technology, and to document analysis, evaluation, and/or imagination of state-of-the-art signals and information technology in a written report;
  - Collaborate and communicate effectively in a team setting.

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Course Description

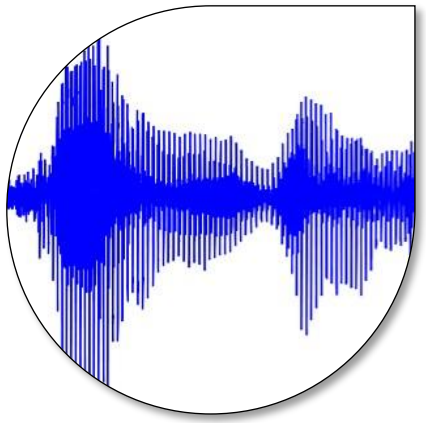
- This general education course introduces the basics of signals and information technology and their applications to daily-life consumer communication devices.
- No prerequisite
- ***BIG IDEAS*** include:
  - the representation of ***signals*** in the time and frequency domains;
  - ***digitization*** of information;
  - ***coding*** for data compression and error protection;
  - ***transmission*** of signals;
  - ***cellular*** networks and
  - ***multiple access*** technologies

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Course Outline

- The Course consists of 6 chapters

Signal and Systems



Digital Systems



Digitization



Coding of Digital Information



Communications



Networks



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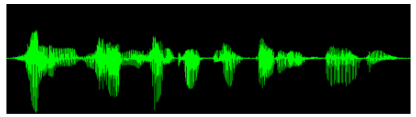
Course Outline

Huawei phone is a digital system



Input signal

What's the weather like today?



Chapter 1: Signals and Systems

How to represent a signal?

What components are contained in a signal?

Chapter 2: Digital Systems

Represent signals in 0 and 1.



Chapter 3: Digitization

How to convert a continuous waveform to a sequence of 0 and 1.



Chapter 4: Coding of Digital Information

How to reduce information size?

How to protect information?

```
00101110011111110110001011 0
01001000101011011101100001 0
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Bring your sunglasses, it's going to be nice in Paris.

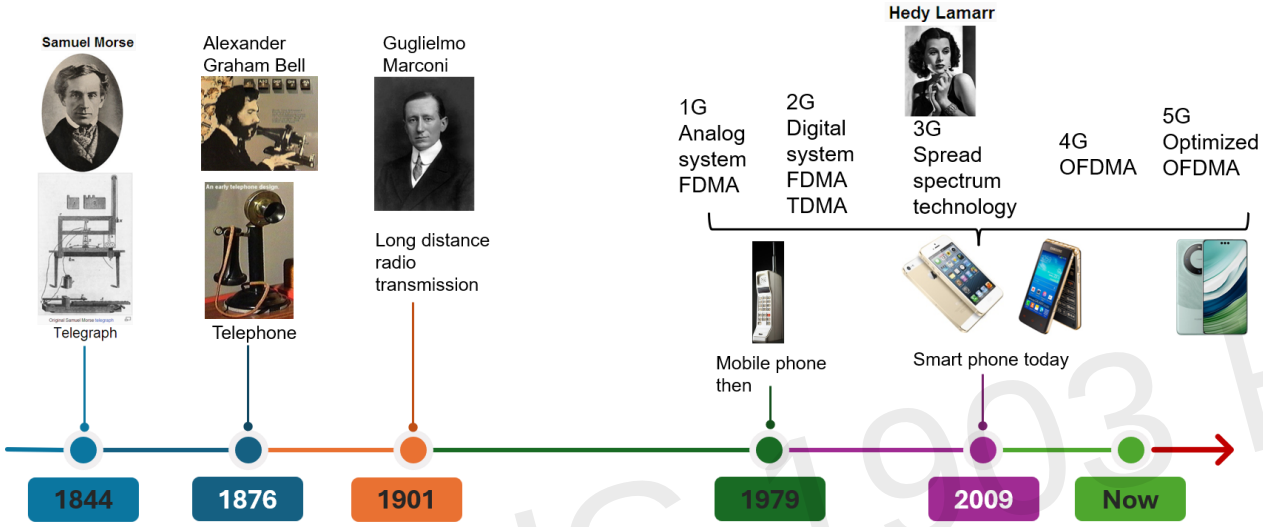
Matching with Phone's database

Output signal

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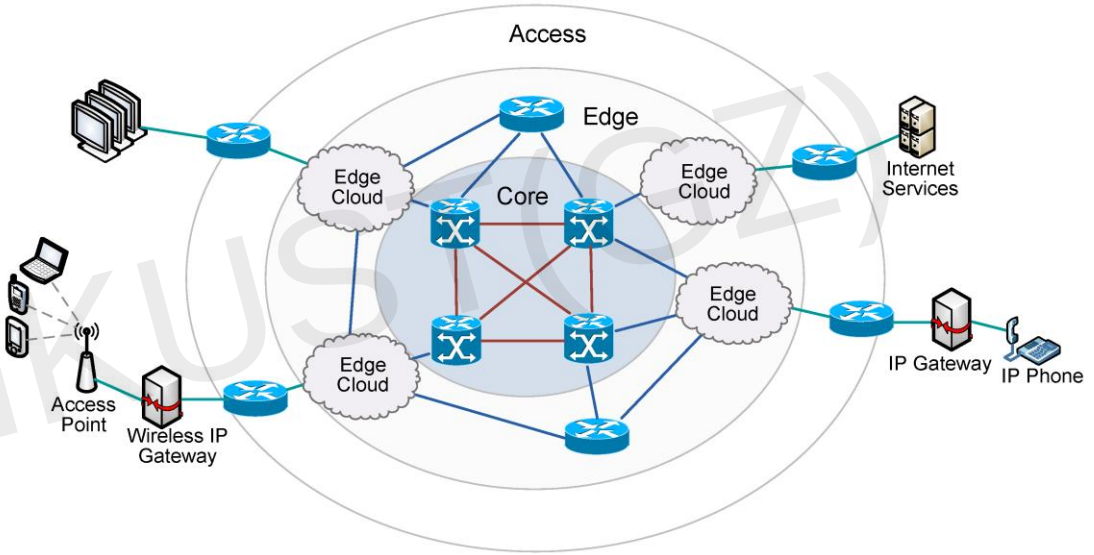
Course Outline

Chapter 5 Communications



- How are phone signals transmitted?
- AM (Amplitude Modulation) and FM (Frequency Modulation)
- Why multiple mobile users can make phone calls in the same cell at the same time?

Chapter 6 Networks



- What's the Internet?
- What's the Internet Protocol?
- Why we need content distribution networks (CDNs)?

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Teaching and Learning Method

Before lectures

The teaching team

Release course materials:
- Lecture notes
- Online articles
- Online videos

Students

- Learn online materials

During Lecture

- Further explanations to concepts and theorems
- Quiz
- Examples
- Demonstration
- In-class activities
  - Discussions
  - Worksheet

Exam

- Midterm
- Final

Project

- Presentation
- Report

Why blended learning?
- Provides personalized instruction
- Reinforces engagement, learning and retention
- Improved student learning outcomes

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Grading

Assignment (20%)	Mid Term (25%)	Final Exam (35%)	Group Project (20%)

❑ **Assignment: 20%**

- Complete the homework simulation tasks guided by the teaching assistants.

❑ **Midterm exam: 25%**

(Multiple choice, fill in blanks, computation and analytical problems, Closed-book exam, no AI)

❑ **Final exam: 35%**

(Multiple choice, fill in blanks, computation and analytical problems, Closed-book exam, no AI)

❑ **Group project: 20%**

- Presentation: 10% (ppt slides + oral presentation)
- Report: 10%

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Team Project

❑ Guidelines

- A group of **2~3 members**. All members of one group will get **the same** project score. Each group to propose one information technology topic based on their interest.
- Project presentation slides (around 10-15 pages)
- Project presentation **in the last week**
- A project report (**maximum1000 words**, to document their views and project findings)
- *Find your teammates as soon as possible and prepare in advance.*

❑ Suggested information technology topics:

- Signal processing technologies (e.g., object/audio/image recognition)
- Wi-Fi (e.g., Wi-Fi 6)
- Satellite communications (e.g., satellite call)
- Smart systems (e.g., smart city, smart scan)
- Optical interconnects
- …

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Team Project Presentation Marking Scheme

PPT Slides

Element \ Score	3	2	1	0
Language and Visual Aid	The language is clear and correct. All the visual aids are clear and legible, and relevant to the content.	The language is clear and correct. Most of the visual aids are clear and legible, and relevant to the content.	The language is mostly clear and correct. Few of the visual aids are clear and legible, and relevant to the content.	The language is not clear and correct. All the visual aids are not clear and irrelevant to the content.
Structure and Organization	All the slides have good layout. The total number of slides is 10-15.	Most of the slides have good layout. Or the total number of slides is not 10-15.	Few of the slides have good layout. And the total number of slides is not 10-15.	The layout of the slides is poor. And the total number of slides is not 10-15.
Content	The content covers all the relevant techniques and background. The contents are presented in a logical order.	The content covers most of the relevant techniques and background. Most of the contents are presented in a logical order.	The content covers few of the relevant techniques and background. Few of the contents are presented in a logical order.	The content does not cover the relevant techniques and background. The contents are not presented in a logical order.
Oral Presentation	The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language.	The presenter achieved any two of the following elements: The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language	The presenter achieved any one of the following elements: The project is clearly presented. The presenter engaged with the audience. The presenter spoke at an appropriate volume, with appropriate body language	The project is not clearly presented. The presenter did not engage with the audience. The presenter did not speak at an appropriate volume, without appropriate body language.

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Team Project Report Marking Scheme

Element \ Score	3	2	1	0
Language and Visual Aid	The language is clear and correct. All the visual aids are clear and legible, and relevant to the content.	The language is clear and correct. Most of the visual aids are clear and legible, and relevant to the content.	The language is mostly clear and correct. Few of the visual aids are clear and legible, and relevant to the content.	The language is not clear and correct. All the visual aids are not clear and irrelevant to the content.
Structure and organization	The report is well organized with the introduction, the main body, and the summary sections.	The introduction or the summary section is missing.	Both of the introduction and the summary sections are missing. Or, the main body section is missing.	The report only contains the introduction or the conclusion section.
Content	The content covers all the relevant techniques and background. The content are presented in a logical order.	The content covers most of the relevant techniques and background. Most of the content are presented in a logical order.	The content covers few of the relevant techniques and background. Few of the content are presented in a logical order.	The content does not cover the relevant techniques and background. The content are not presented in a logical order.
Citation Note: for AI-generated citation, the writer should follow the format below and append the screen shot of the question and AI-generated response to the question. 1. In-text citation: ([Citation number]) 2. Reference list: [Citation number] ChatGPT, response to author query. OpenAI [Online]. https://chatgpt.pro/ (accessed February 15, 2023)	The writing meticulously adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, consistently and accurately citing all sources used in the text and references section. In-text citations follow the proper IEEE format, including author names, publication years, and page numbers when necessary. The references section is comprehensive and properly formatted, with all required elements, such as author names, publication dates, titles, and source information, presented in accordance with IEEE guidelines.	The writing consistently adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, accurately citing sources used in the text and references section. In-text citations generally follow the proper IEEE format, including author names and publication years. The references section is comprehensive and properly formatted, with all required elements, such as author names, publication dates, titles, and source information, mostly presented in accordance with IEEE guidelines.	The writing partially adheres to the Institute for Electrical and Electronics Engineers (IEEE) citation style, with occasional errors or inconsistencies in the citation format. In-text citations may not consistently follow the proper IEEE format for author names, publication years, or page numbers. The references section may have some omissions or inaccuracies in the required elements, such as author names, publication dates, titles, or source information.	The writing does not adhere to the Institute for Electrical and Electronics Engineers (IEEE) citation style, with numerous errors or inconsistencies in the citation format. In-text citations do not follow the proper IEEE format for author names, publication years, or page numbers. The references section may be incomplete, inaccurately formatted, or missing required elements, such as author names, publication dates, titles, or source information.

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Signals and Information Technology Matters

- This course in Signal and Information Technology lays a crucial foundation for any major you may choose in the future.

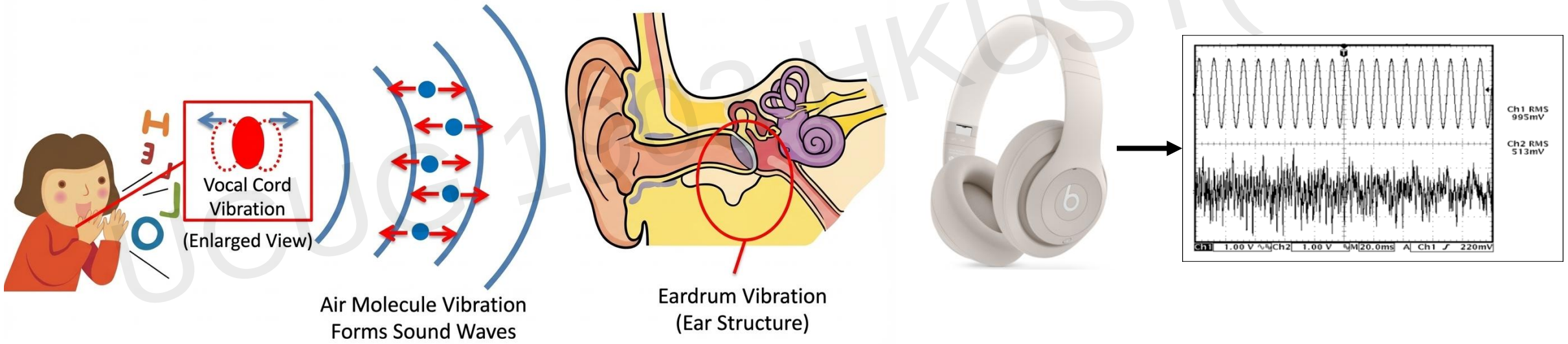


Program Name
BEng in Materials Science and Engineering
BEng in Microelectronics Science and Engineering
BEng in New Energy Science and Engineering
BEng in Artificial Intelligence
BSc in Data Science and Big Data Technology
Extended Major Program in Artificial Intelligence
Minor Program in Artificial Intelligence
BEng in Robotics
BEng in Smart Manufacturing Engineering
BSc in Financial Technology

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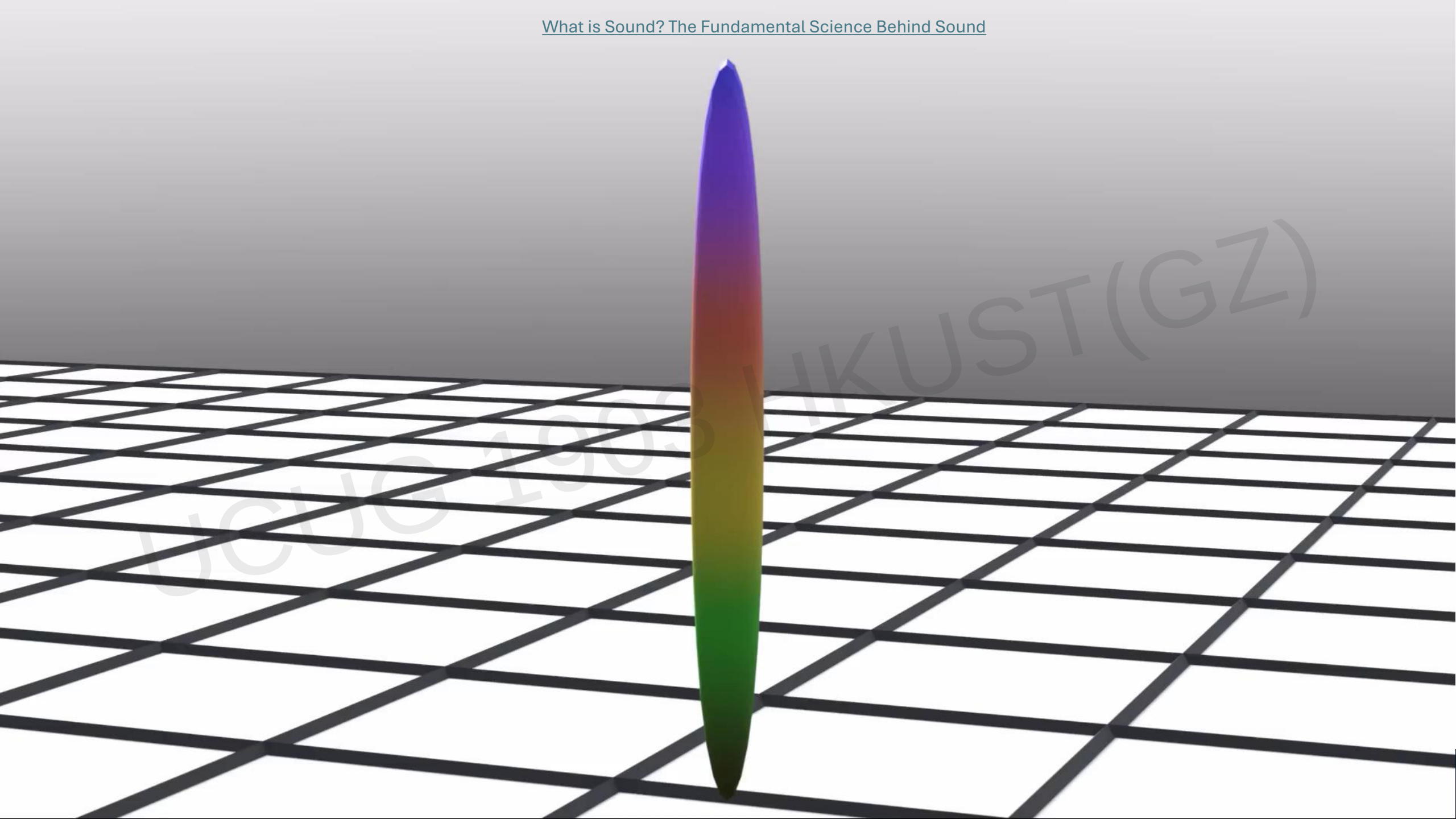
What is Signal?

- Signal: A pattern or variation that conveys information.
- A signal can be what we see, hear, touch, smell, taste. Examples: Sound wave, Light wave.
- Signals in engineering: audio, image, video



Pictures are from https://ce.hkfyg.org.hk/wp-content/uploads/sites/110/2018/04/sound1.jpg and modified by nano banana

What is Sound? The Fundamental Science Behind Sound



Signal is Everywhere

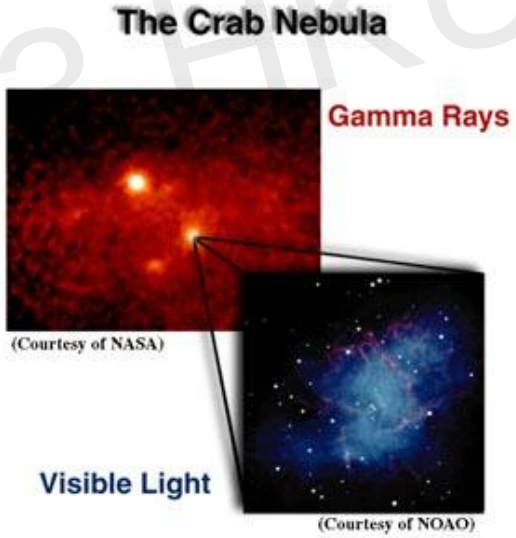
- A signal can be contained in what we do not sense directly

Radio, cellular phone, and wireless LAN signal

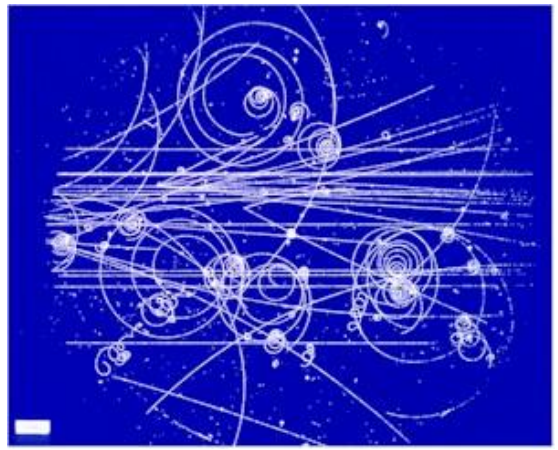


Signals in science (Physics)

Gamma ray from the universe



High-energy particle tracks



Pictures are from https://e.huawei.com/en/products/wlan, https://www.iotallknow.com/cellular-technology/

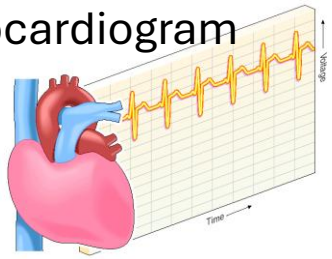
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Signal Conveys Information

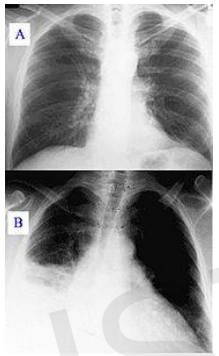
Shanghai Stock Index

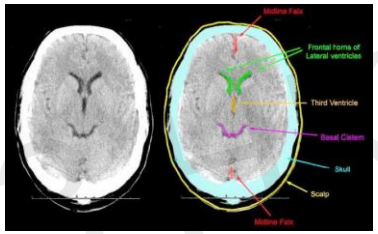


Electrocardiogram



X-ray





Computer Tomography





Signals carry information.

Signals may bring stimulations and enjoyment.

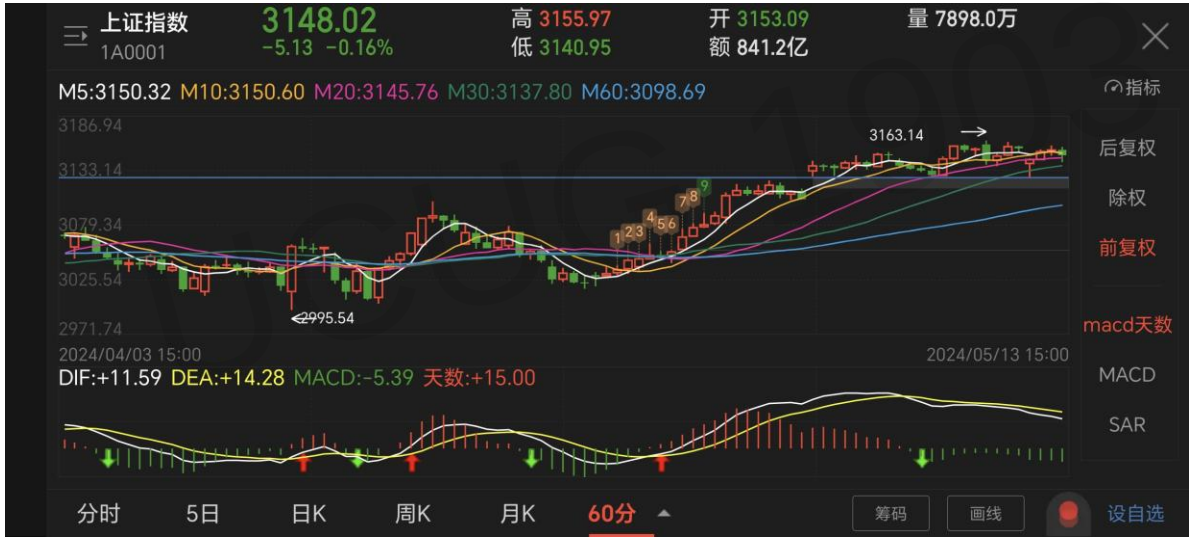
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Representation of Signal

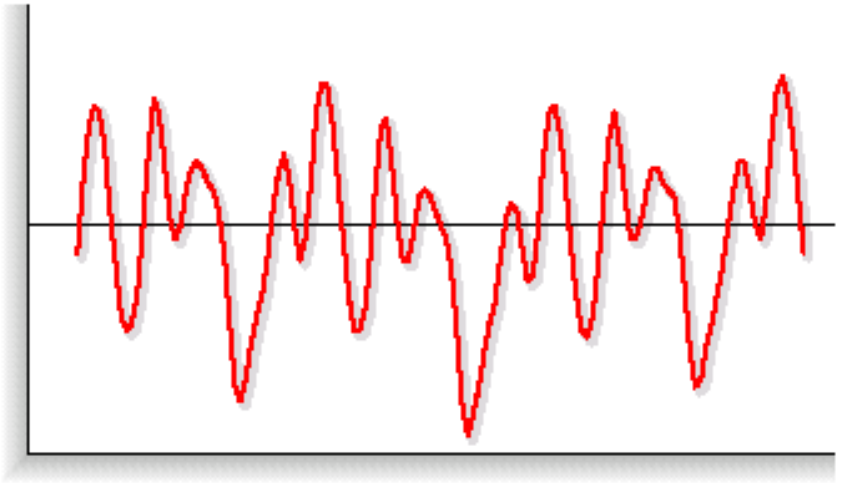
- Often, signal is the variation of some physical quantity over ***one or more independent variables.***

- Common independent variables: time t, spatial coordinates (x,y), or both (x,y,t)

Variation of the Shanghai Stock Index over one month



Sound Signal – Variation of Acoustic Pressure over 1/20 of a second

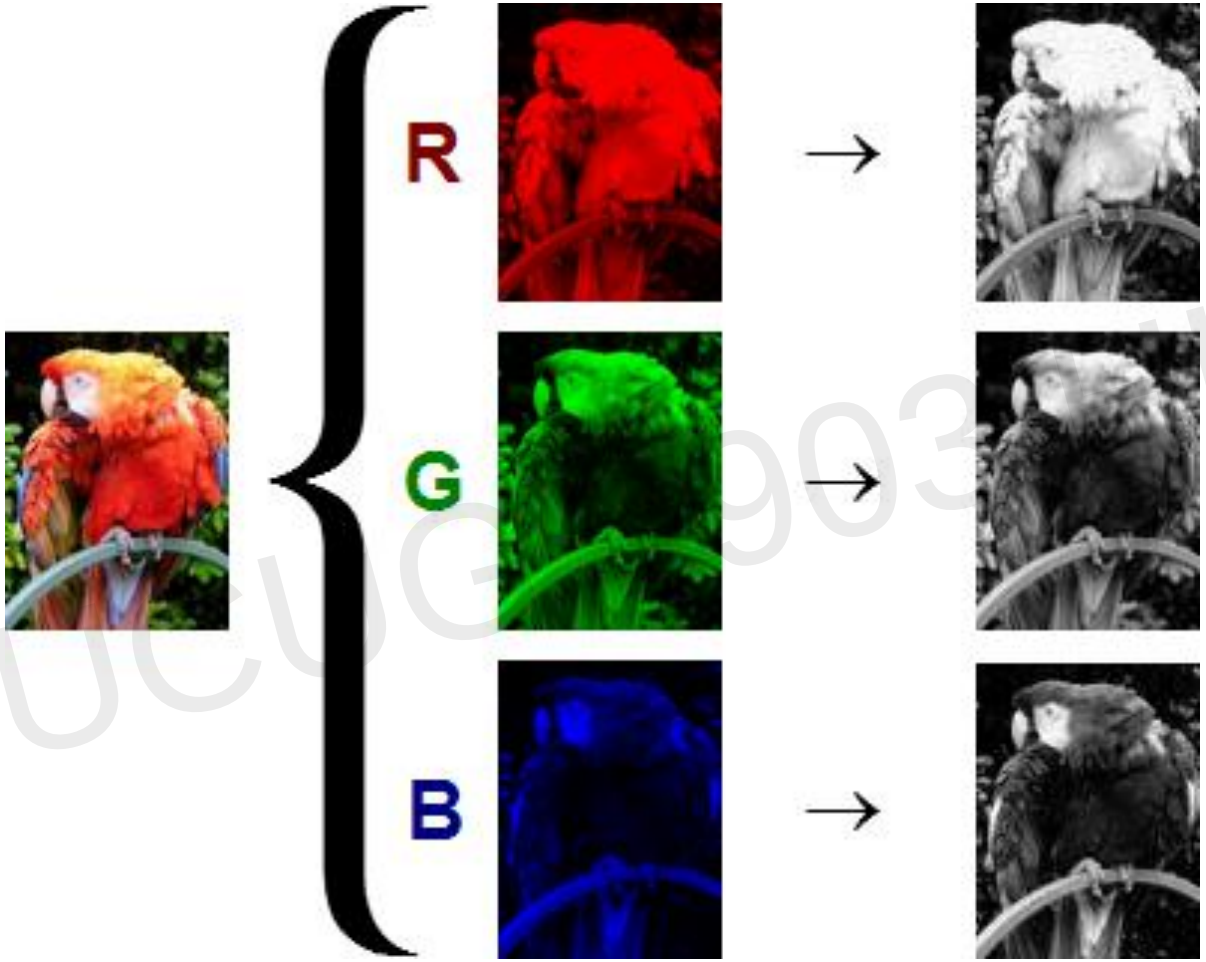


50 msec

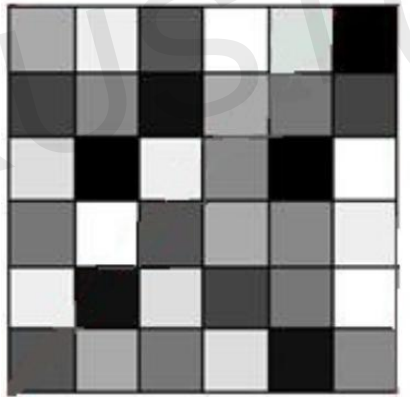
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Representation of Signal

- Signal can be variations of some physical quantity over ***space.***



https://upload.wikimedia.org/wikipedia/commons/5/56/RGB_channels_separation.png



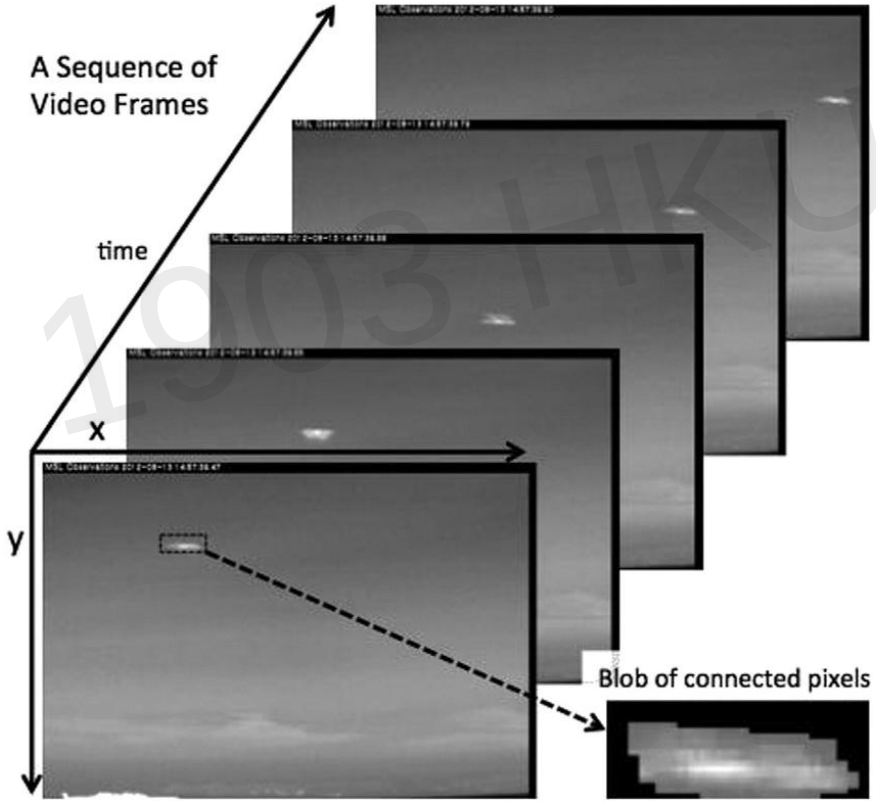
170	238	85	255	221	0
68	136	17	170	119	68
221	0	238	136	0	255
119	255	85	170	136	238
238	17	221	68	119	255
85	170	119	221	17	136

https://www.cnblogs.com/ChanYanny/p/18475117

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Representation of Signal

- Signal can also be variations over *both space and time*. Example: video
- But in engineering, it is often very important for us to think of signals as functions of ***frequency***.
  (to be discussed in lecture 3)



https://www.researchgate.net/figure/A-video-is-a-sequence-of-images-called-frames-Each-frame-is-a-two-dimensional-grid-of_fig2_281719202

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Classification of Signals

- Analog vs. Digital Signals

  Whether the signal values are hold an infinite range of values or discrete levels.

- Periodic vs. Aperiodic Signals

  Whether the signal repeats itself at regular intervals.

- Energy Signals vs. Power Signals

  Whether the signal's total energy or average power is finite.

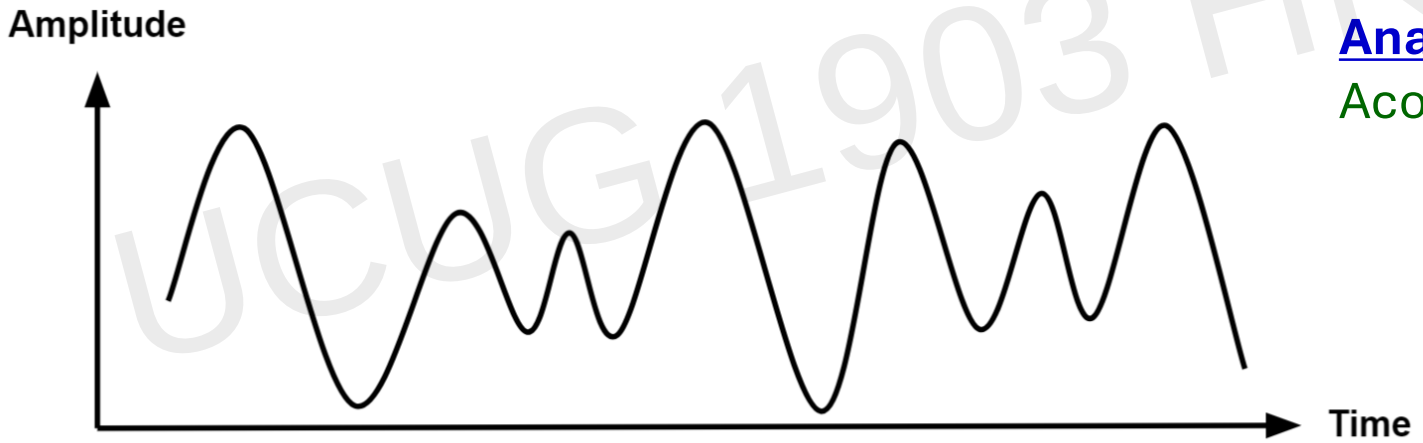
- Deterministic vs. Random Signals

  Whether the future values of the signal can be predicted precisely.

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Analog Signals

- ***Analog signals*** are signals that vary continuously over time/space with arbitrary continuous values.
- Many real-life quantities such as length, weight, force, acoustic pressure, electrical current, etc., are viewed as analog signals.

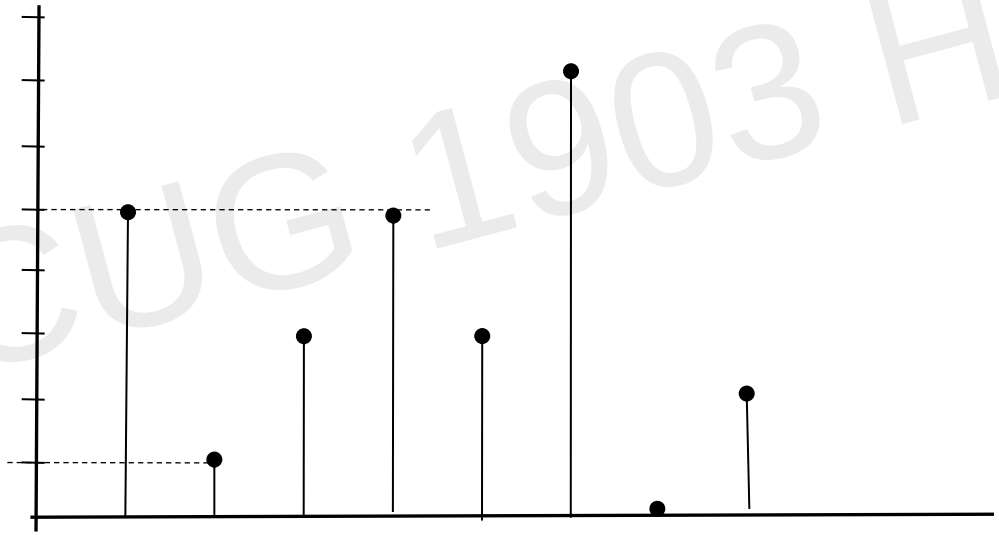


Analog Signal –
Acoustic Pressure; Electrical Current

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Digital Signals

- ***Digital signals*** are signals that are defined only at discrete time/space instances and can take on a finite set of values.
- All the signals processed by modern computers are processed as digital signals.



- **Digital Signal –**

Student attendance at lecture

Shanghai Stock Index at end of day

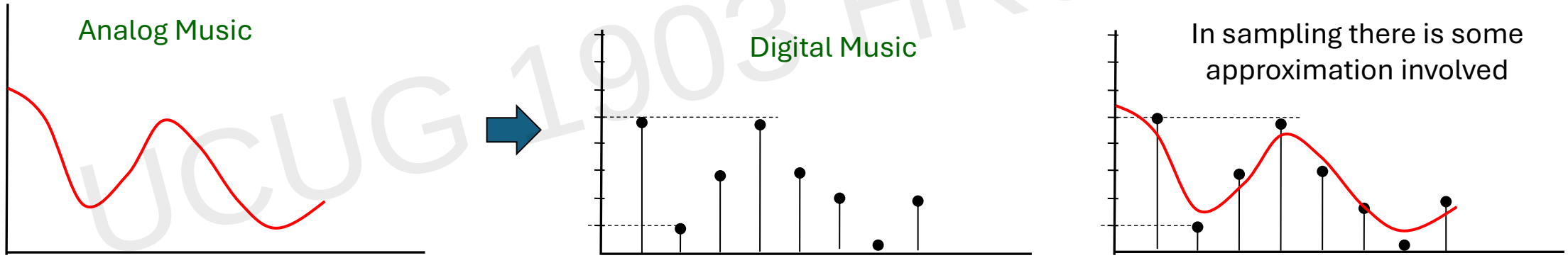
MP3 Music file

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Conversion Between Analog and Digital Signals

❑ We frequently convert between analog and digital signals (e.g. to process analog audio signals by the computer).

Example: Sampling of Analog Music Signal for storage in a Digital CD

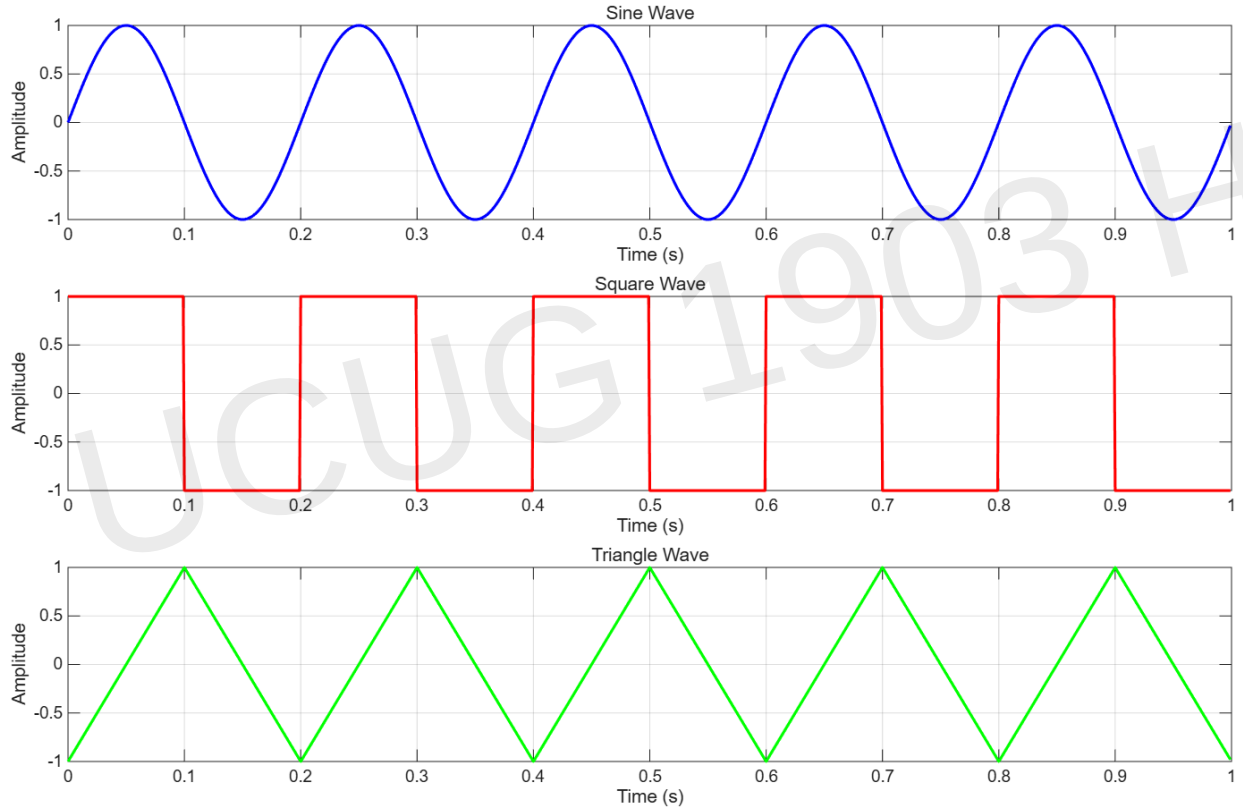


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Periodic Signal

- ***Periodic Signal*** are signals that repeats itself at regular intervals over time or space.

- They model repetitive phenomena and serve as the basis for analyzing and processing all types of signals through techniques like Fourier analysis.



Periodic Signal –

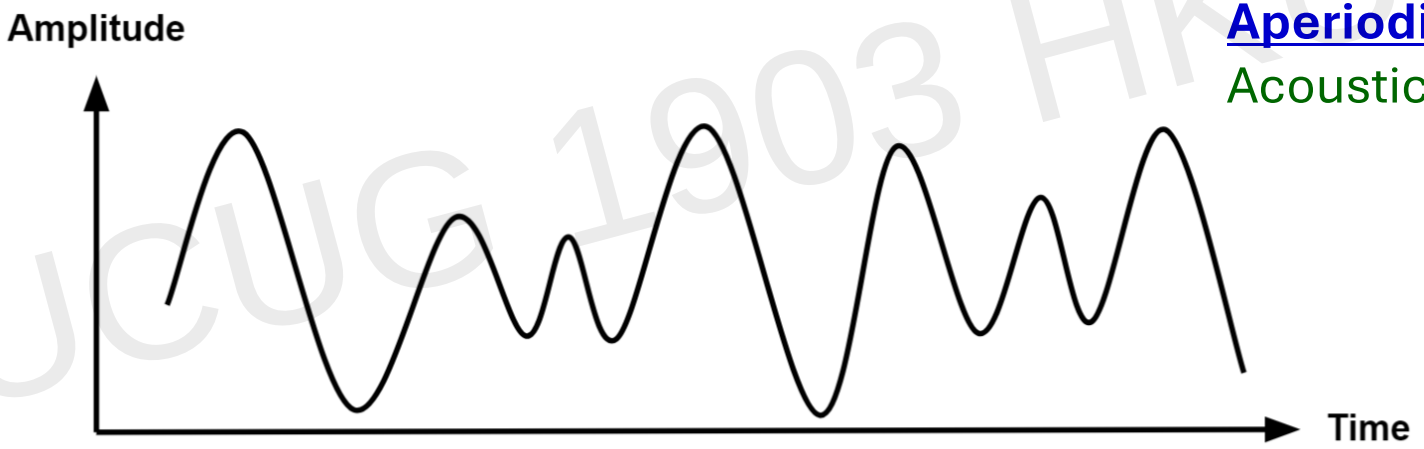
Sine wave, Square Wave, Triangular Wave

$$f(t) = f(t + T)$$

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Aperiodic (Non-Periodic) Signal

- ***Aperiodic Signals*** are signals that do not repeat over time or space
- They are closer to real-life signals and are the primary focus of signal processing

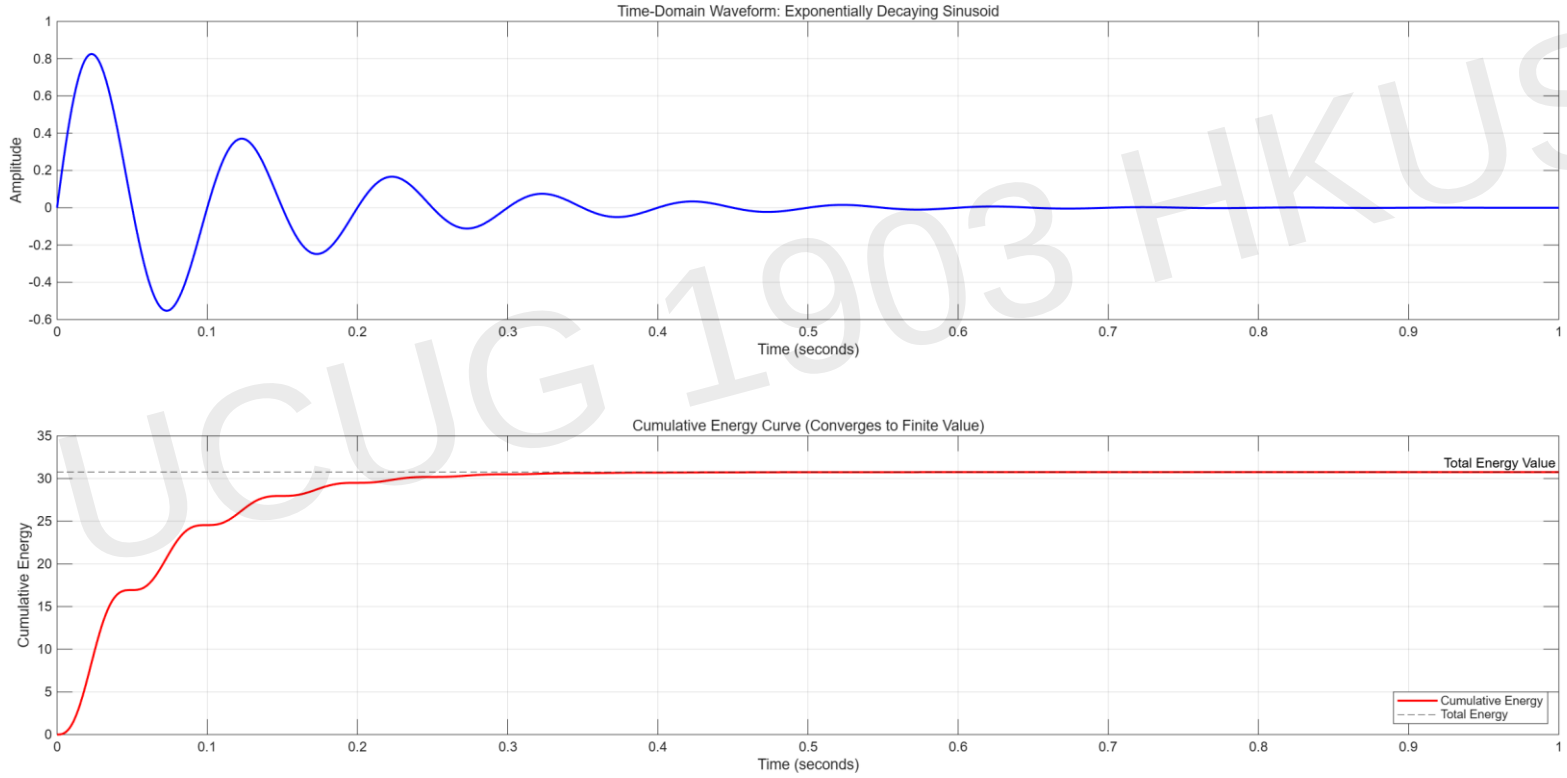


Aperiodic Signal –
Acoustic Pressure

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Energy Signal

- ***Energy signal*** are signals with finite total energy and zero average power.
- Many real-world transient phenomena, such as a single light pulse, a damped echo, or a sudden burst of sound, are viewed as energy signals.

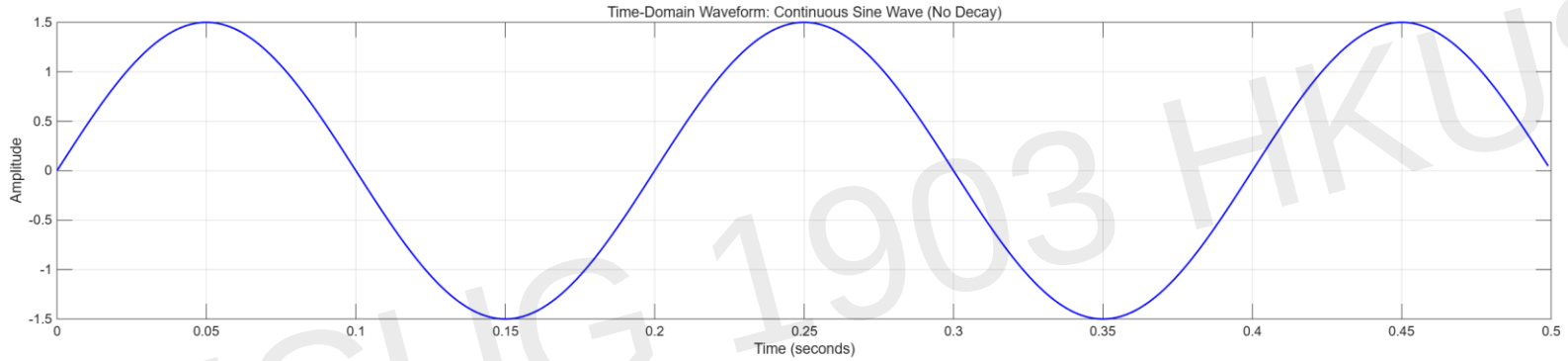


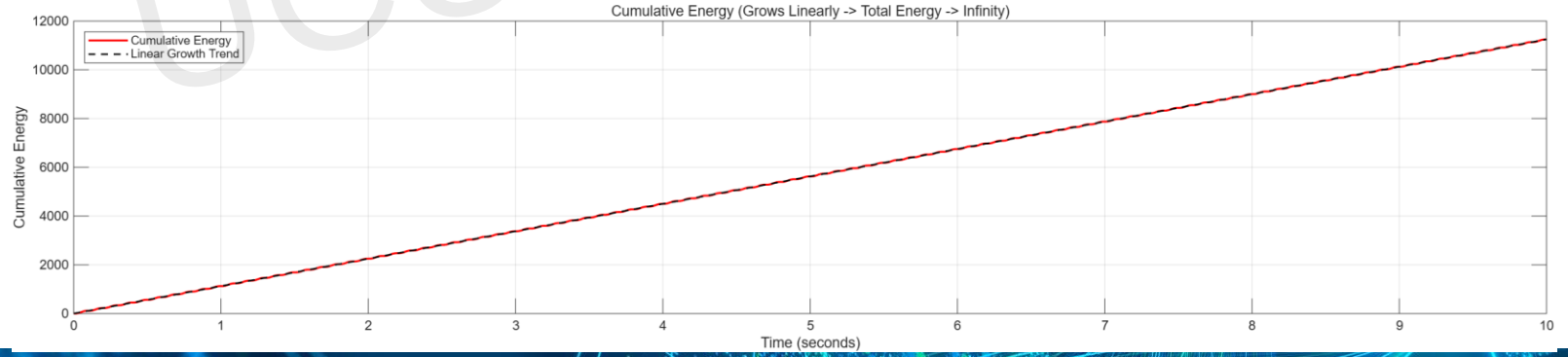
Energy Signal – Exponentially Decaying Sinusoid

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Power Signals

- ***Power signals*** are signals with finite, non-zero average power and infinite total energy.
- Many continuous or periodic phenomena, such as a steady sine wave, the constant hum of an air conditioner, or the continuous flow of electrical current in a power line, are viewed as power signals.



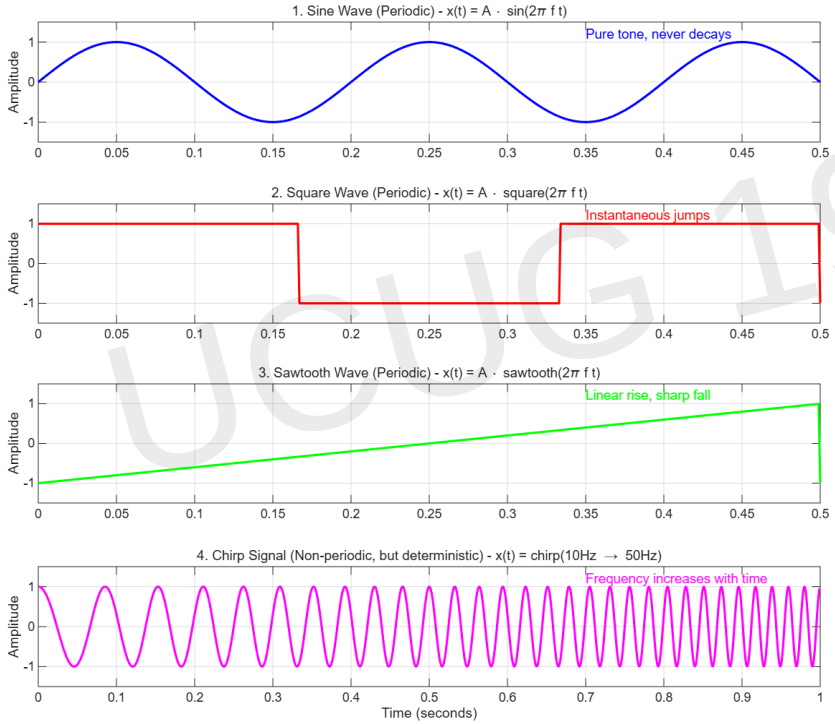


Power Signal –
Pure Sinusoid

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Deterministic Signals

- ***Deterministic signals*** are signals that can be precisely described by a mathematical expression or a specific rule, with no uncertainty about their future values.
- Many engineered signals, such as a sine wave generated by an oscillator, the carrier wave in radio transmission, or a step function in control systems, are viewed as deterministic signals.



$$x_1(t) = A \cdot \sin(2\pi f t + \phi)$$

$$x_2(t) = \begin{cases} A, & 0 < t < \frac{T}{2} \\ -A, & \frac{T}{2} < t < T \end{cases}, \quad x_2(t+T) = x_2(t)$$

$$x_3(t) = 2A\left(\frac{t}{T} - \left\lfloor \frac{t}{T} + \frac{1}{2} \right\rfloor\right)$$

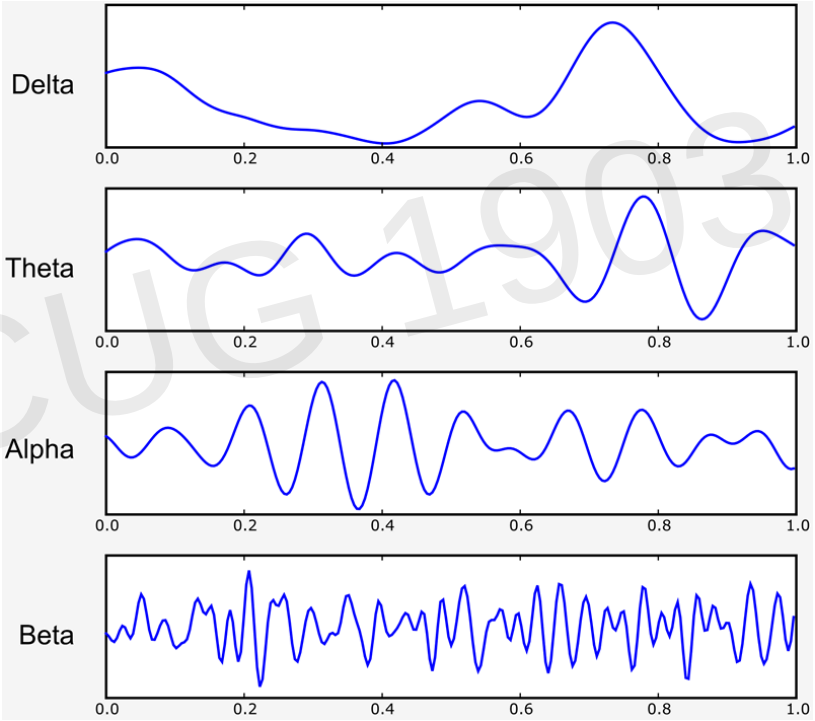
$$x_4(t) = A \cdot \sin\left(2\pi\left(f_0 t + \frac{f_1 - f_0}{2T} t^2\right)\right)$$

Deterministic Signal –
Sine Wave, Square Wave,
Sawtooth Wave,
Chirp Signal.

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Random Signals

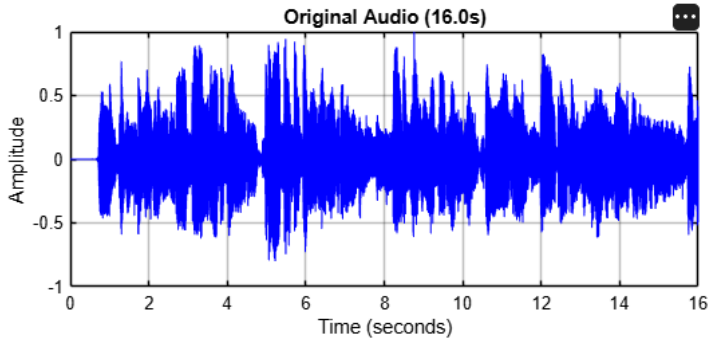
- ***Random signals*** are signals that cannot be predicted exactly and must be described using probability and statistics, exhibiting inherent uncertainty.
- Many real-world natural or complex phenomena, such as background noise in a recording, the fluctuations in stock market prices, or the electrical activity of the human brain (EEG), are viewed as random signals.

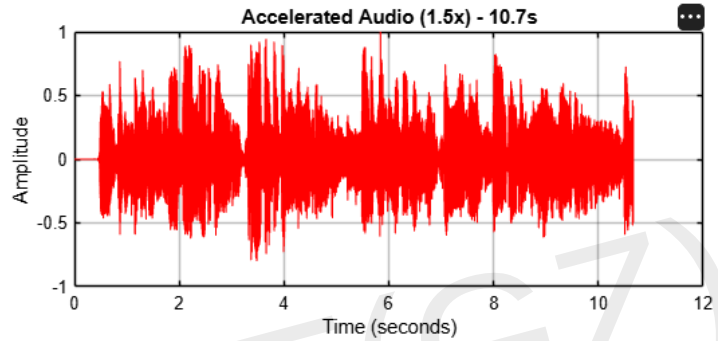


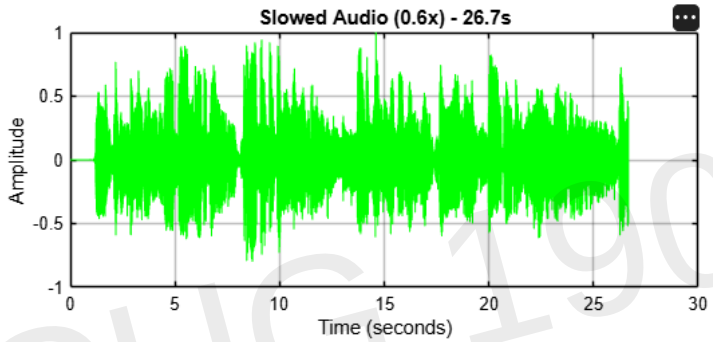
Random Signal –
EEG signal, Acoustic Pressure

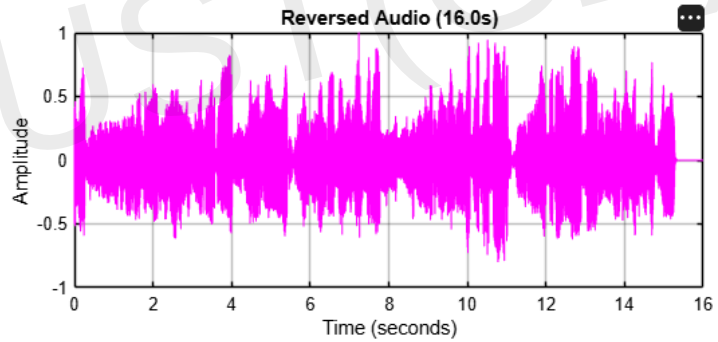
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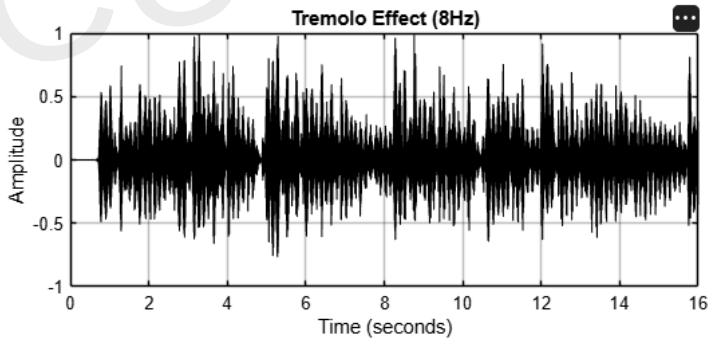
In-Class Activity – A Simple Example on Signal Processing

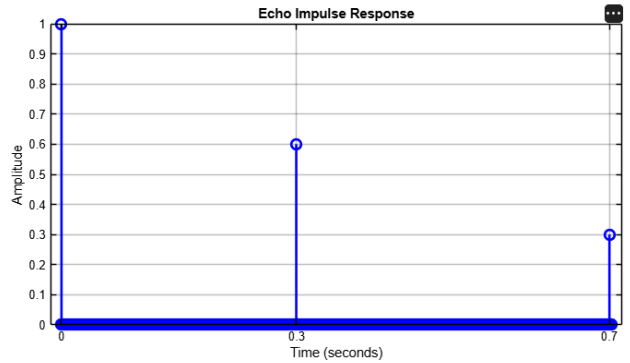












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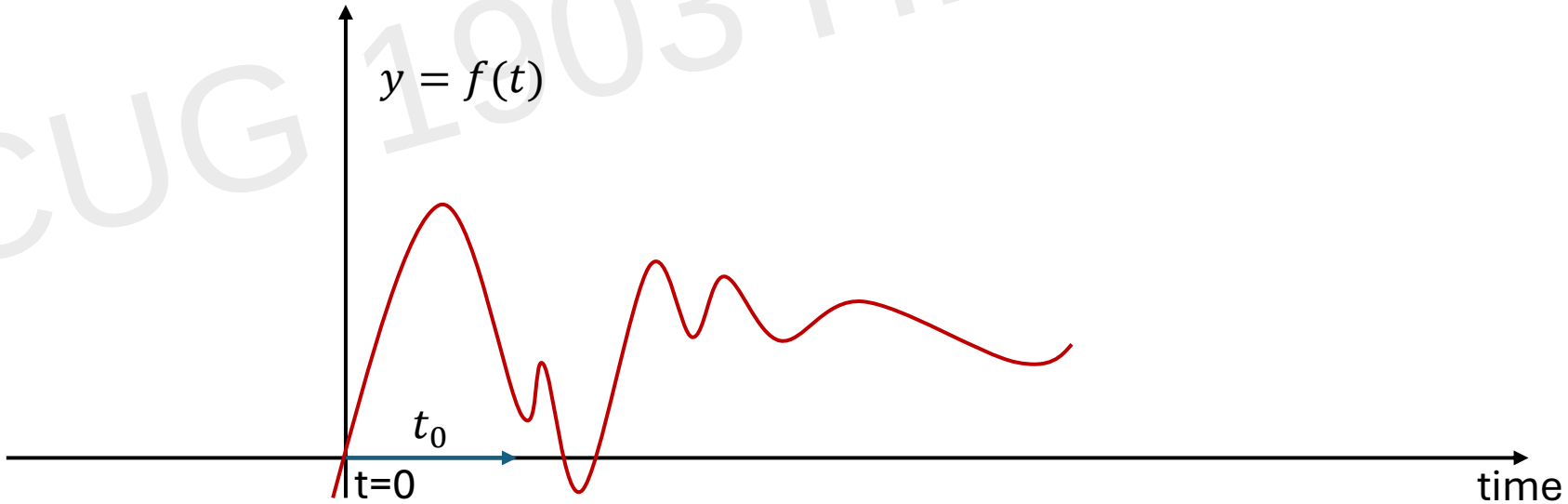
Operations on Signals

- ❑ Shift
- ❑ Scale
- ❑ Invert
- ❑ Addition
- ❑ Multiplication

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Shift

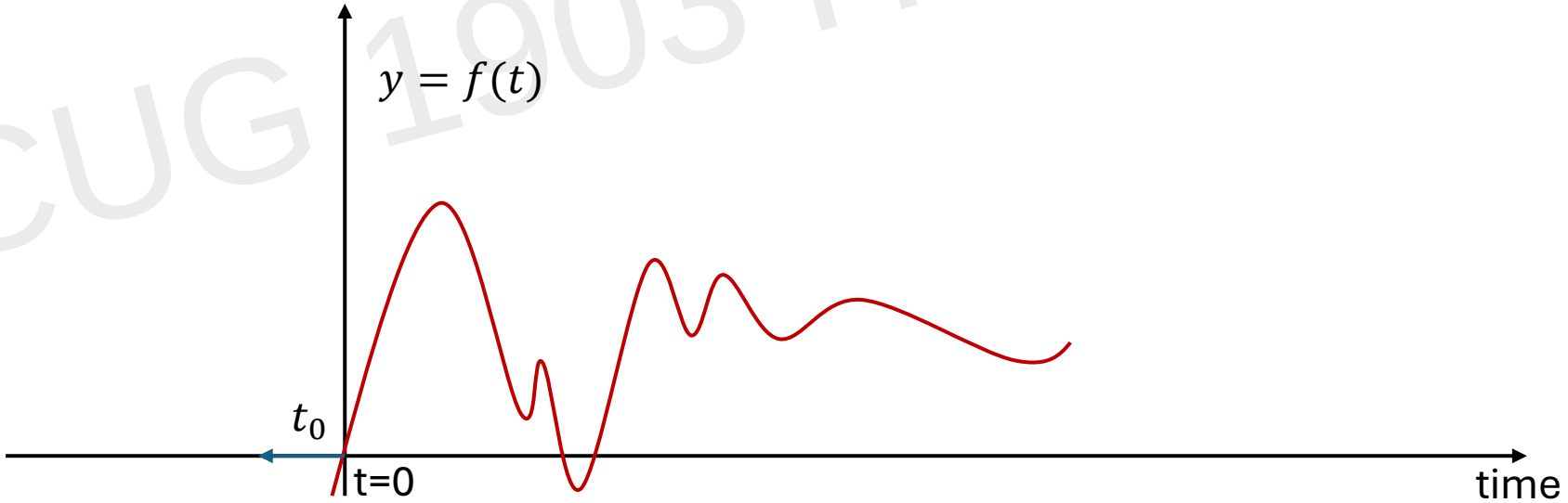
- **Time Shifting (or Signal Shift)** refers to the operation of moving a signal's waveform horizontally to the left or right along the time axis (the horizontal axis), while **keeping the shape of the waveform completely unchanged**.
- For $f(t - t_0)$: when $t_0 > 0$, the waveform is shifted to the right by t_0 seconds. This operation is called Delay.



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Shift

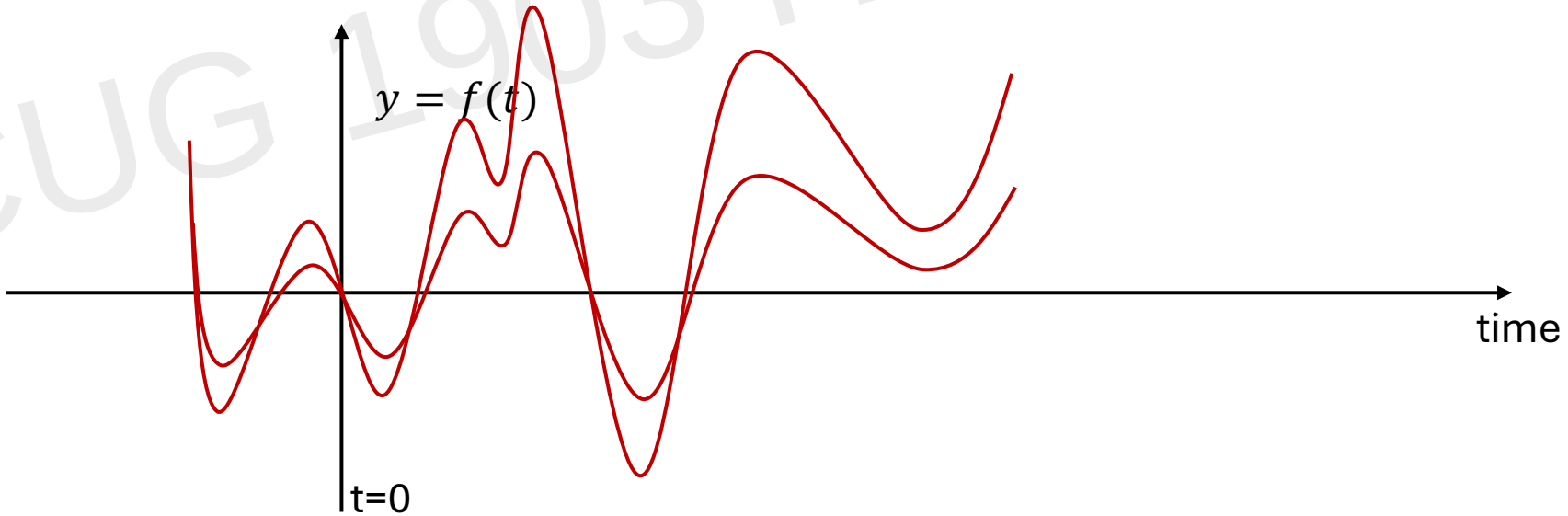
- **Time Shifting (or Signal Shift)** refers to the operation of moving a signal's waveform horizontally to the left or right along the time axis (the horizontal axis), while **keeping the shape of the waveform completely unchanged**.
- For $f(t - t_0)$: when $t_0 > 0$, the waveform is shifted to the right by t_0 seconds. This operation is called Delay.
- For $f(t + t_0)$: when $t_0 > 0$, the waveform is shifted to the left by t_0 seconds. This operation is called Advance (or Time Advance).



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Scale

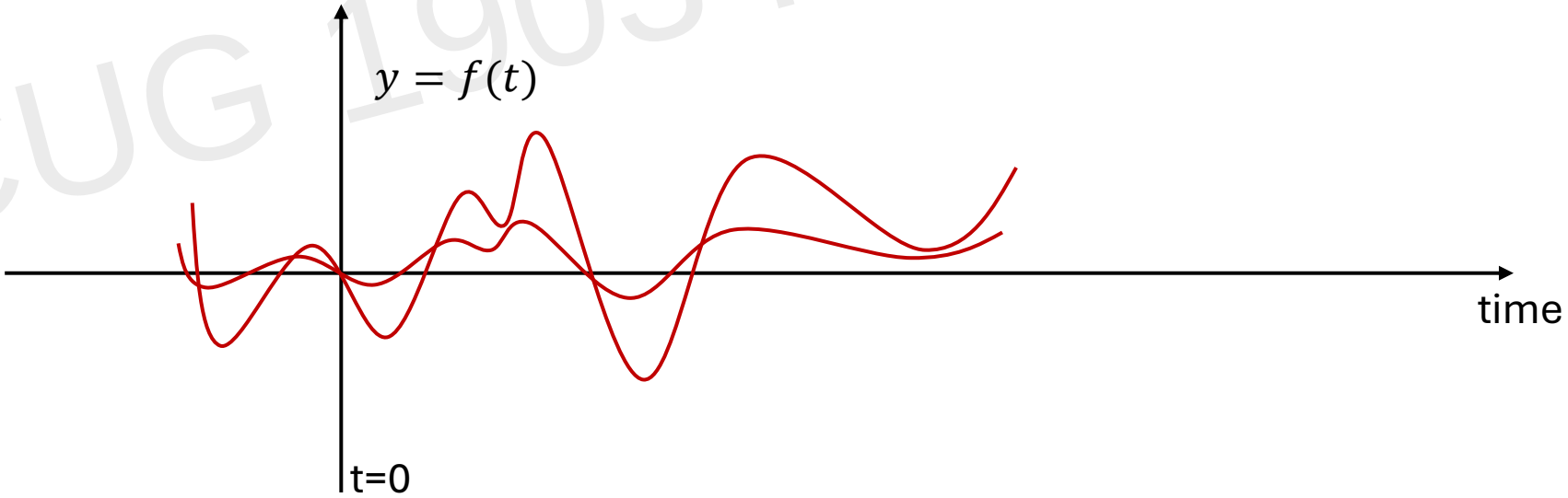
- **Signal Scaling** generally refers to stretching or compressing the horizontal axis (time axis) or the vertical axis (amplitude axis) of a signal.
  - **Amplitude Scaling:** $y(t) = A \cdot f(t)$
    - When $A > 1$: the waveform becomes **taller** (amplitude increases).



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Scale

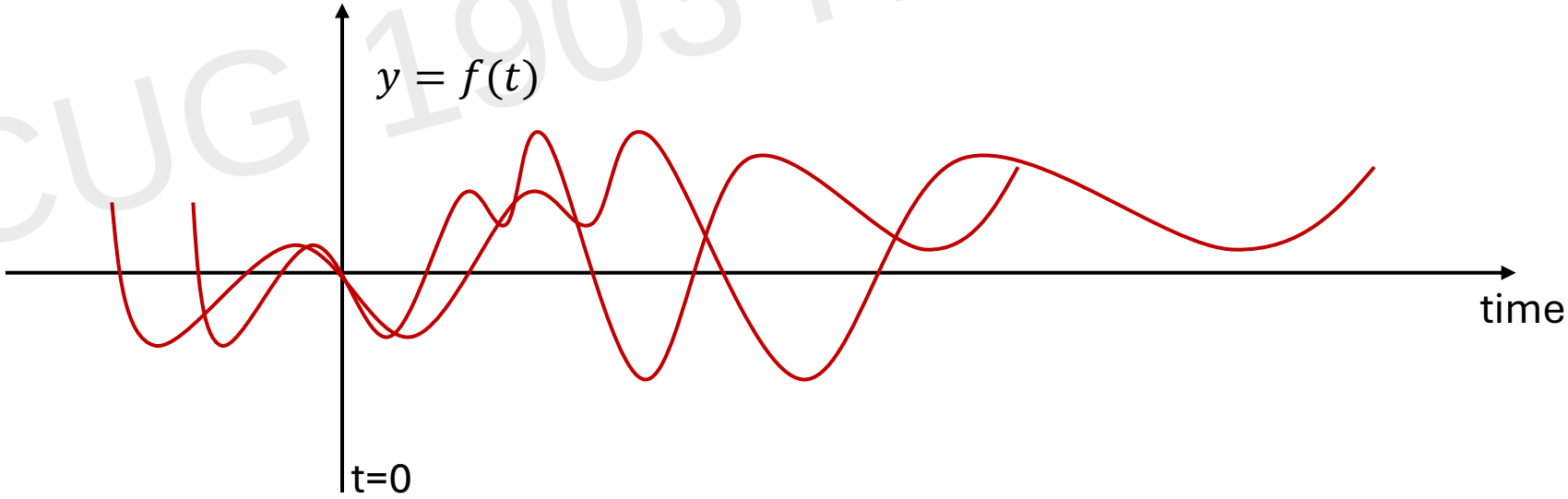
- **Signal Scaling** generally refers to stretching or compressing the horizontal axis (time axis) or the vertical axis (amplitude axis) of a signal.
  - **Amplitude Scaling:** $y(t) = A \cdot f(t)$
    - When $A > 1$: the waveform becomes **taller** (amplitude increases).
    - When $0 < A < 1$: the waveform becomes **shorter** (amplitude decreases).



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Scale

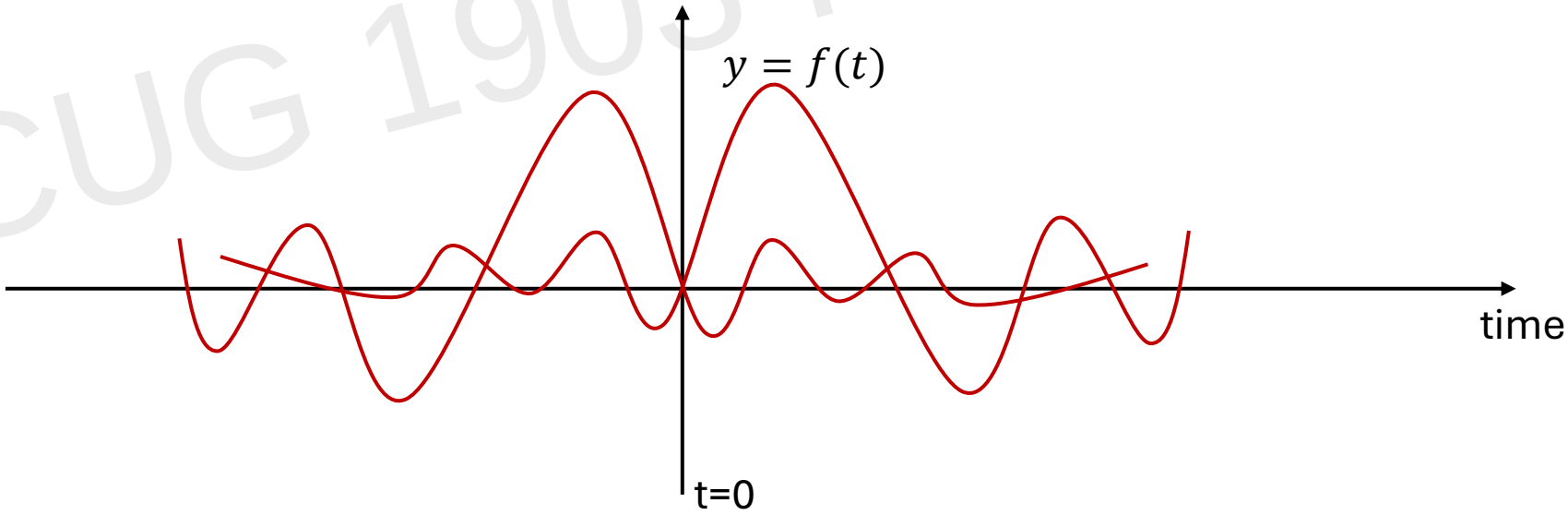
- **Signal Scaling** generally refers to stretching or compressing the horizontal axis (time axis) or the vertical axis (amplitude axis) of a signal.
  - **Amplitude Scaling:** $y(t) = A \cdot f(t)$
  - **Time Scaling:** $y(t) = f(a \cdot t)$
    - When $0 < a < 1$: the waveform becomes **wider (stretched)**. The signal is "stretched" in time, meaning it changes **slower** (lower frequency).



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Invert

- There are two types of signal inversion in signal processing
  - **Time Reversal (Folding) – Horizontal Flip:** $y(t) = f(-t)$

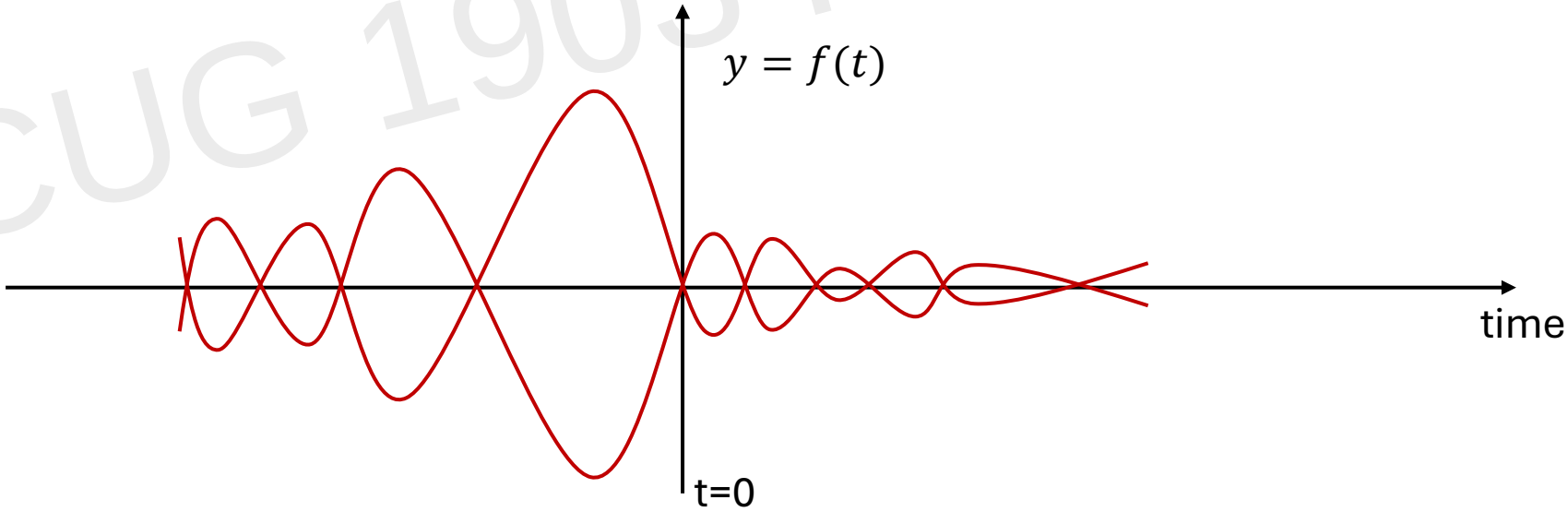


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Invert

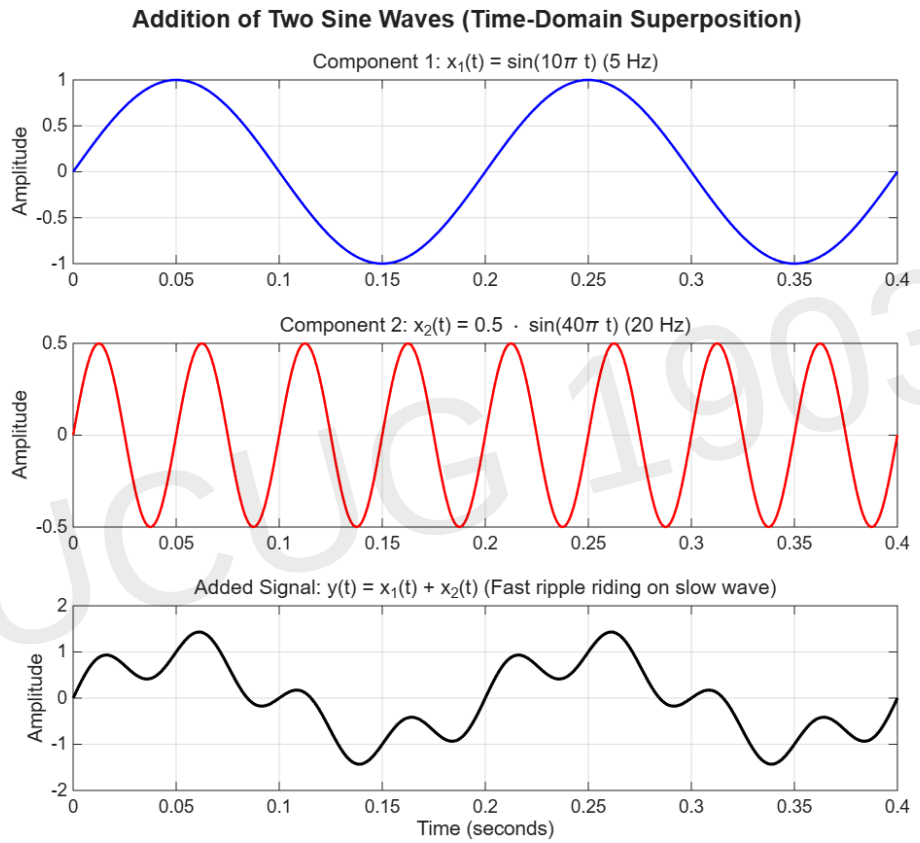
- There are two types of signal inversion in signal processing
  - **Time Reversal (Folding) – Horizontal Flip:** $y(t) = f(-t)$
  - **Amplitude Inversion (Polarity Reversal) – Vertical Flip:** $y(t) = -f(t)$

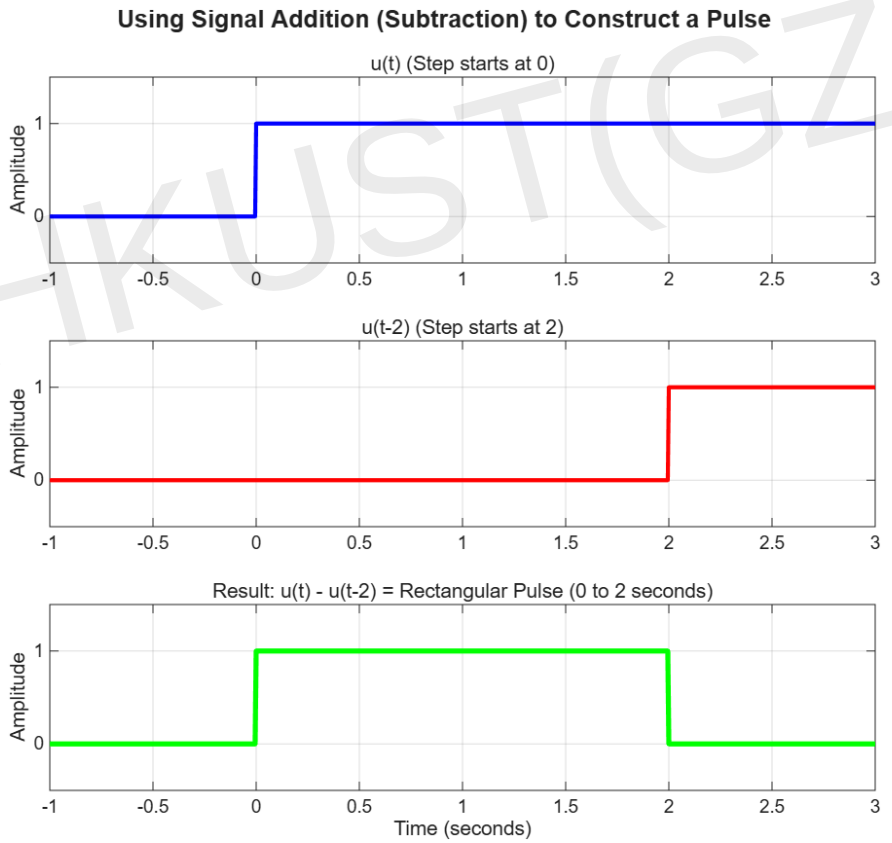
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Addition

- **Signal Addition**, mathematically speaking, is the operation of **directly adding the amplitudes of two or more signals at the same point in time** to generate a new composite signal.

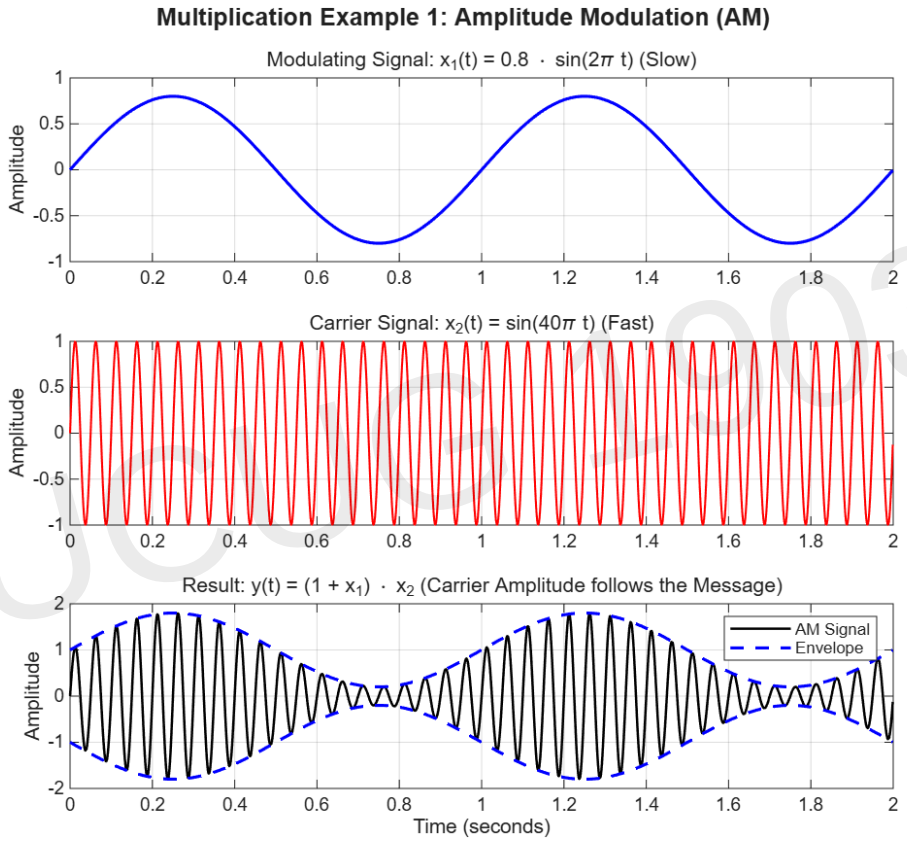


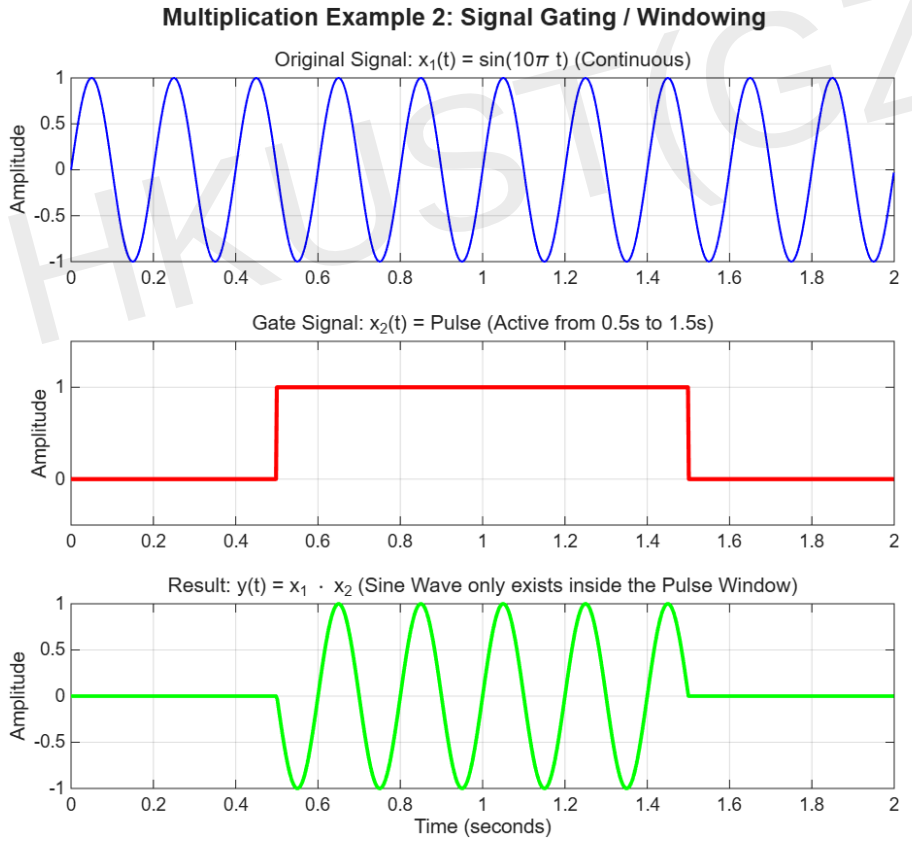


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Multiplication

- **Signal Multiplication**, in mathematical terms, refers to the **point-by-point multiplication of the amplitude values of two signals at the same time instant**, thereby generating a new signal.

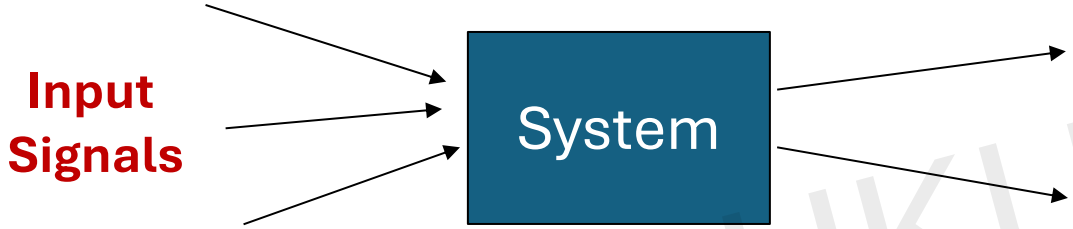




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What is System

- In scientific and engineering studies, everything can be modeled as **systems** with signals as input and signals as output

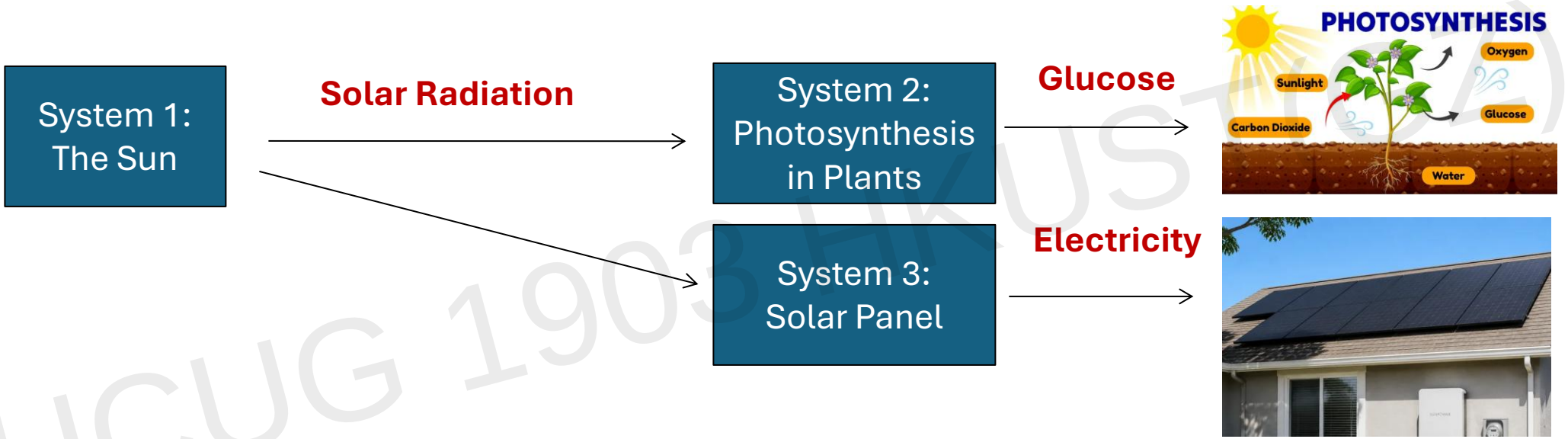


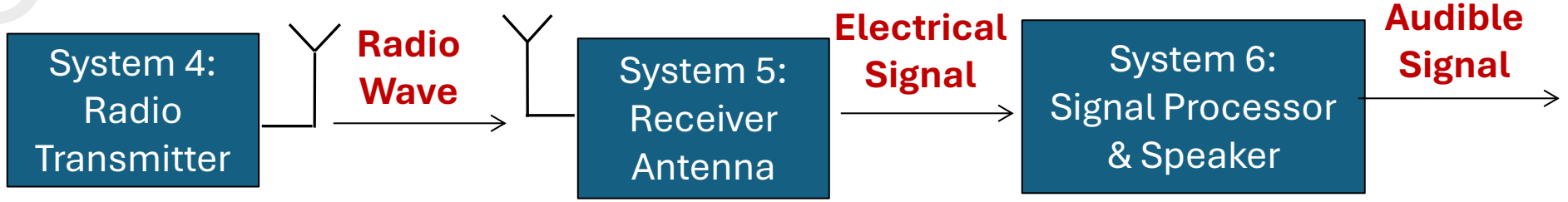
Input	System	Output
Deposit/withdrawal, interest rate	Bank account	Account balance
Acoustic signal	Human auditory	Perception of sound
Radio signal	Huawei phone	Audio signal
Light signal	Digital camera	Digital image

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Signals as Input and Output of Systems

- Output signals from one system is often the input to another system.
- Signals transfer information from one system to another system and bring interactions among systems.



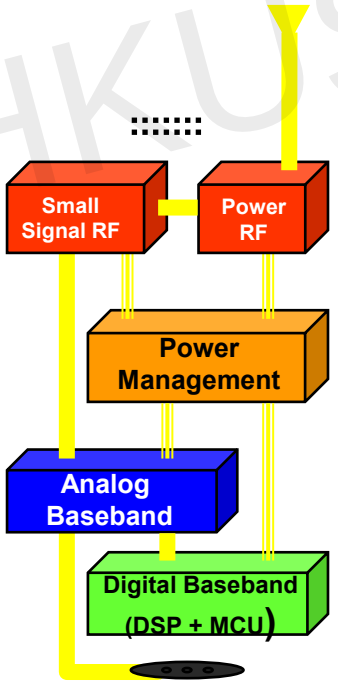


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Cell Phone – A Complex System in Your Hand

- A system can be complex!
- **E.g.** The cell phone is a system that contains several sub-systems (i.e. systems within a system), and many sub-sub-systems within the sub-systems. The cell phone represents one of the most complex systems ever invented (yet so compact, multi-functional and popular!).





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Signal Is Only Useful When We Can Manipulate It

- ***Common techniques to manipulate various kinds of signal include:***
  - Analyze
  - Convert
  - Modify
  - Compress
- Most of the consumer electronic products actually perform various kinds of signal processing, e.g.

Phone  → Radio Signal Processing

MP3 player  → Audio Signal Processing

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Analyze: Decoding the Signal's Secrets

- Analysis is the first step in signal processing. It involves observing the signal to extract meaningful patterns or data without necessarily changing the signal itself.
  - **Spectrum Analysis:** Breaking a complex signal into its individual frequencies (e.g., seeing the bass vs. treble in a song).
  - **Feature Extraction:** Identifying specific markers, like a heartbeat rhythm in an EKG or a face in a video feed.
  - **Noise Identification:** Determining what part is useful data and what is unwanted interference.



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Convert: Bridging the Physical and Digital Worlds

- Signals often need to change their fundamental state to be useful to different systems.
  - **Analog-to-Digital (ADC):** Taking continuous physical waves (like sound) and turning them into 1s and 0s that a computer can read.
  - **Digital-to-Analog (DAC):** Turning stored digital data back into physical waves so we can hear music through a speaker.
  - **Domain Conversion:** Moving a signal from the **Time Domain** to the **Frequency Domain** (often using the Fourier Transform) to make it easier to work with.



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Modify: Shaping the Signal for the User

- Modification changes the actual properties of the signal to achieve a specific goal, such as improving clarity or adding effects.
  - **Filtering:** Removing unwanted noise (High-pass, Low-pass, or Band-pass filters).
  - **Amplification:** Increasing the signal's strength or volume.
  - **Equalization (EQ):** Adjusting specific frequency ranges to balance the output.
  - **Modulation:** Overlaying a signal onto a carrier wave for long-distance radio or Wi-Fi transmission.



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Compress: Maximizing Efficiency and Speed

- Compression reduces the amount of data required to represent a signal, which is essential for streaming and storage.
  - **Lossless Compression:** Reducing file size without losing a single bit of data (e.g., ZIP files or FLAC audio).
  - **Lossy Compression:** Removing "unnecessary" information that the human eye or ear can’t perceive to save massive amounts of space (e.g., MP3 or JPEG).
  - **Why it matters:** Without compression, streaming a 4K movie would require a bandwidth that most home internet connections couldn't handle.

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Property of System

- ❑ Linearity
- ❑ Time-Invariance
- ❑ Causality
- ❑ Stability

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Linearity

- A system is classified as linear if and only if it satisfies Homogeneity (Scaling) and Additivity (Superposition).

  Homogeneity (Scaling): $T[a \cdot x(t)] = a \cdot T[x(t)]$

  Additivity (Superposition): $T[x_1(t) + x_2(t)] = T[x_1(t)] + T[x_2(t)]$

- Suppose a system T maps an input signal $x(t)$ to an output $y(t)$, denoted as $T[x(t)]=y(t)$.

- For the system to be linear, for any constants a_1, a_2 and any inputs $x_1(t), x_2(t)$, it must satisfy:

$$T[a_1x_1(t) + a_2x_2(t)] = a_1T[x_1(t)] + a_2T[x_2(t)] = a_1y_1(t) + a_2y_2(t)$$



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Time-Invariance

- Time-Invariance means that the system's behavior (its parameters) does not change over time.
- Let a system T map an input signal $x(t)$ to an output $y(t)$:

$$T[x(t)] = y(t)$$

- The system is **time-invariant** if for **any real number** t_0 (the time shift), the following holds:

$$T[x(t - t_0)] = y(t - t_0)$$

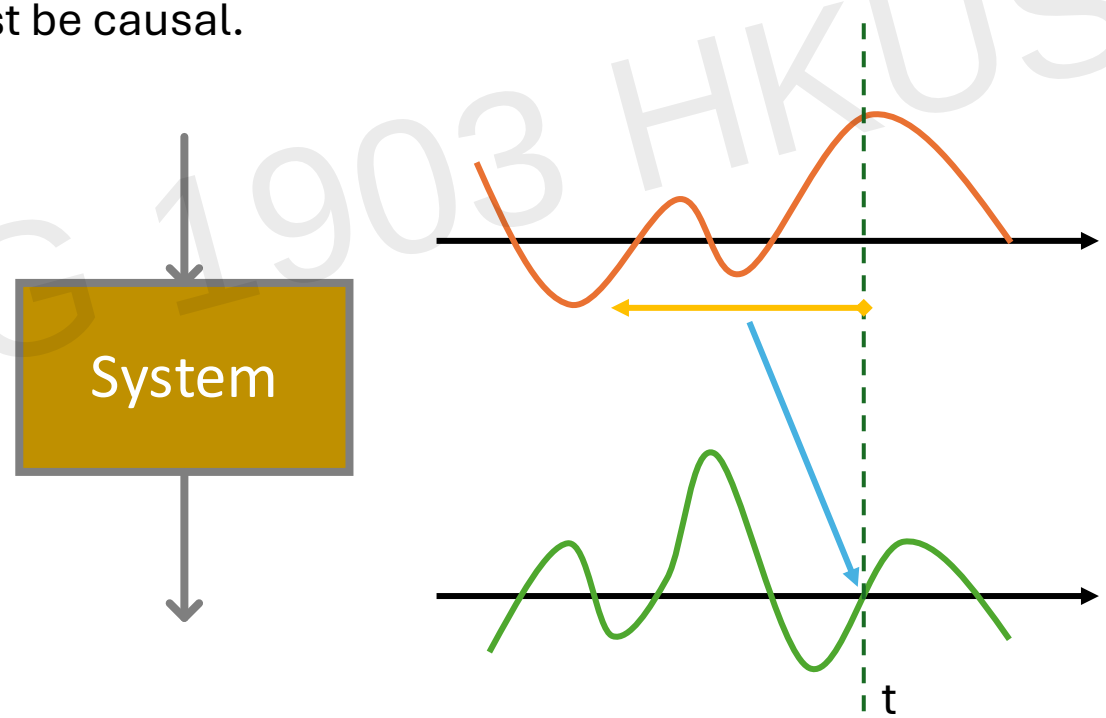
- **Interpretation**: If you delay the input by t_0 seconds, the output is simply the original output delayed by the same t_0 seconds. The shape and amplitude of the waveform remain unchanged.



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Causality

- A system is causal if and only if the output at any time depends only on the current and past inputs, and never on future inputs.
  - **Continuous-time system**: The output $y(t)$ at time t depends only on the input $x(\tau)$ for $\tau \leq t$.
  - **Discrete-time system**: The output $y[n]$ depends only on $x[n], x[n-1], x[n-2], \ldots$.
- A real-time system must be causal.

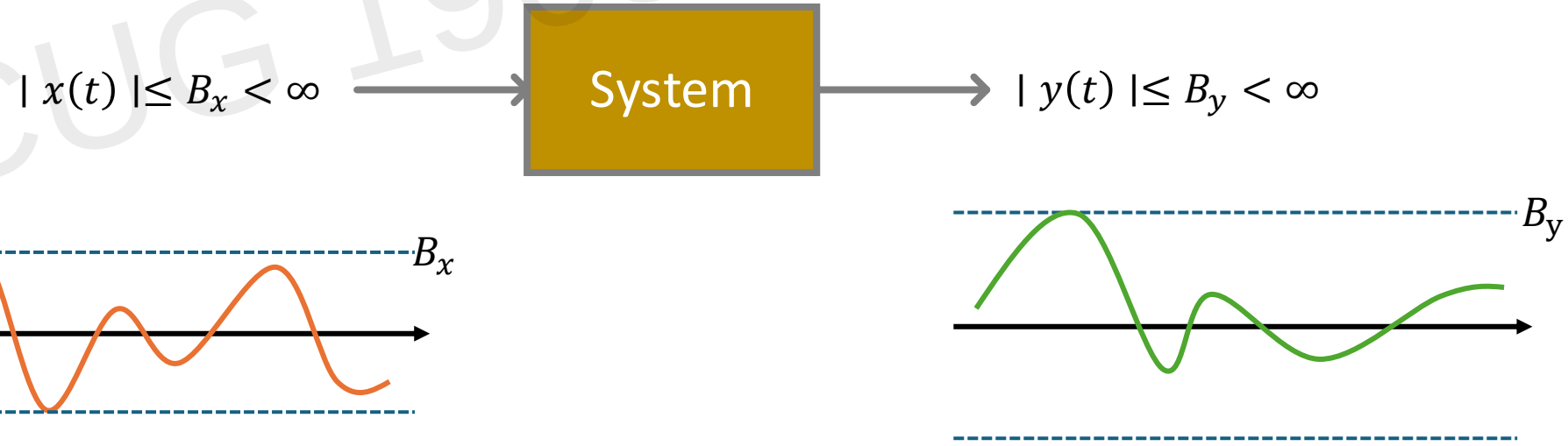


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Stability

- **Stability** is a fundamental property that determines whether a system's output remains "well-behaved" (bounded) when subjected to bounded inputs. The most common and important criterion for stability in signals and systems is **BIBO Stability** (Bounded-Input Bounded-Output Stability).

- **A system is BIBO stable if and only if every bounded input produces a bounded output.**
  - A bounded input means there exists a finite constant B_x such that $|x(t)| \leq B_x < \infty$ for all t.
  - A bounded output means there exists a finite constant B_y such that $|y(t)| \leq B_y < \infty$ for all t.



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Some Engineering Notations (Prefix)

- In engineering and science we often encounter dimensions that can be very large or very small. So let us introduce some engineering prefixes for convenience of our future technical discussions

Prefix	Symbol	Value
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
(no prefix)		10^{0}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}

- Examples: “4 GB” USB flash memory, “10M Pixels Digital Camera”, “1 Gbps broadband service.”
- You have probably heard about “nanotechnology.” “Nano,” in fact, is an engineering notation for 10^{-9}.

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Summary

- We discussed **analog** and **digital signals** and **systems**.
- In Lecture 2&3, we focus on analog signals and introduce two fundamental concepts for the understanding of signals and their processing by systems – ***spectrum*** and ***filtering***.
- In Lecture 4&5, we will discuss digitalization of signals and how digital signals are processed. Digitalization is the foundation of the modern information age.
- Next in Lecture 2, we start by looking at the sound signal as an example.

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